

New Evidence of the Usefulness of Real-Time Sentiment Indices in Econometric and Judgement Based Forecasting Models*

Kaiwen Qiu^a, Ali Kabiri^{b,c}, John S. Landon-Lane^a, Norman R. Swanson^a

^a *Rutgers University*

^b *Buckingham University*

^c *Center for the Study of Decision-Making Uncertainty, University College London*

September 2026

Abstract

In this paper, we construct a set of new narrative sentiment indices using textual data from a large corpus of U.S. economic and policy documents. The indices are designed for use in real-time forecasting contexts, and are constructed using transformer-based language models and LLM-assisted annotation. They include: a finance-domain transformer; an economics-tone model trained on LLM-generated annotations of Federal Reserve, Bureau of Labor Statistics, and Census text; and a topic-conditioned narrative measure. In order to determine the usefulness of these three indices for forecasting, we carry out a series of experiments in which their marginal predictive content is assessed. In particular, we first assess the predictive content of the indices in econometric models that are used for real-time U.S. macroeconomic and financial forecasting. We then assess whether our sentiment indices contain information that can be utilized to improve the (quasi) judgemental forecasts reported in the United States Survey of Professional Forecasters (SPF). This is done by testing the rationality of SPF forecasts when conditioning on both real-time macroeconomic variables and our sentiment indices. Our prediction experiments indicate that the new sentiment indices offer substantial marginal predictive content for a variety of variables. We also find evidence consistent with a hypothesis that information contained in the indices is already incorporated in SPF forecasts, as evidenced by the paucity of efficiency rejections in our rationality test experiments. This finding, taken together with our real-time forecasting results underscores the potential usefulness of the new indices.

Keywords: Real-time forecasting, Forecast Rationality and Efficiency, Narrative and Consumer Sentiment, Large Language Models, Text-as-Data.

*Corresponding author: Norman R. Swanson, Department of Economics, Rutgers University, 75 Hamilton Street, New Brunswick, NJ 08901, USA, nswanson@rutgers.edu. Kaiwen Qiu: Department of Economics, Rutgers University, USA, kq60@economics.rutgers.edu. Ali Kabiri: Department of Economics and International Studies, Buckingham University, Hunter Street, Buckingham, MK18 1EG, United Kingdom, ali.kabiri@buckingham.ac.uk. John Landon Lane: Department of Economics, Rutgers University, USA, jlandonl@economics.rutgers.edu. The authors thank Hyunhak Kim, Yang Liu, Kenwin Muang, Shaojun Tong for helpful comments and suggestions on earlier versions of this paper.

1 Introduction

The importance of the interplay between the use of judgement and econometric models when constructing predictions has been discussed in many papers in the economics literature. One common view amongst practitioners, as stated by Lawrence, Goodwin, O’Connor, and Onkal (2006), is that judgement is an “... indispensable component of forecasting ...”. Given that the judgemental component of a forecast is shaped not only by quantitative information sets but also by the narratives and tone present in economic and financial news, it should be of use to investigate a “judgement channel” that is measured with recently developed text-as-data methods. This leads to our approach in this paper of developing new sentiment indices that are extracted from large textual corpora using modern natural language processing (NLP) and large language model (LLM) methods. We evaluate these indices by examining their marginal predictive content when used in both judgement and econometrics model-based forecasts, after conditioning on a real-time macroeconomic dataset, professional forecasts obtained from the Survey of Professional Forecasters, and extant consumer sentiment indices already available in the literature.

Rather than relying on a single textual signal based on NLP and LLM methods when constructing our sentiment indices, we construct three complementary transformer-based narrative sentiment indices. These are collectively referred to as our LLM-based sentiment indices, and differ in the way text is processed: Namely, we construct a finance-domain index (FinBERT); an “economics-tone” index (EconBERT) trained on large-language-model annotations of Federal Reserve, BLS, and Census releases; and a topic-conditioned index (Topic-Fused) that uses zero-shot classification to form soft memberships over pre-specified economic topics, constructs a tone measure within each topic, and then equally weights eight topic-specific tones.

The potential usefulness of sentiment indices for forecasting is discussed in many papers. For example, one literature documents that textual measures of tone, attention, and uncertainty carry predictive content for asset prices and the macro-economy. Tetlock (2007) shows that media pessimism forecasts downward pressure on stock prices, Garcia (2013) documents that the predictive content of news sentiment is concentrated in recessions. Additionally, Baker, Bloom, and Davis (2016) construct a newspaper-based index of economic policy uncertainty that is informative for real activity. More recently, A. H. Shapiro, Sudhof, and Wilson (2022) build a news-based sentiment index for the U.S. economy, Bybee, Kelly, Manela, and Xiu (2024) extract interpretable news-attention factors that track business cycles, and Ke, Kelly, and Xiu (2026) show that supervised text models can predict returns out of sample. Within this tradition, Johnson and Tuckett (2022) examine the role of sentiment as a mediating variable in judgement. They show the link between narratives, which is the interpretation of text by agents, and their subsequent forecast choices in a laboratory setting. Sharpe, Sinha, and Hollrah (2023) apply an emotion-word based lexicon to textual narratives of Federal Reserve ‘Greenbook’ forecasts, which combine model-based and judge-

ment forecasts, and are able to improve economic forecasting outcomes for the U.S. macroeconomy. Sharpe et al. (2023) argue that narrative sentiment is useful for incorporating new information into forecasts. In their framework, as well as ours, sentiment is closely related to the concept of ‘affect’. The notion of ‘affect’ involves the idea that stimuli create an emotional based evaluation, which may be unconscious (see e.g. Slovic, Finucane, Peters, and MacGregor (2002)) and that the ‘emotional core’ of an attitude is the main determinant of many judgments and behaviors that can impact judgement through several potential channels.

The approach taken in this paper draws on an idea from the neuroscience literature that all cognition (knowing) is based on a mixture of cognitive skills and affect (see L. Shapiro (2011)). Moreover, such emotions/ feelings are often useful cues to judgement because they track real information about the environment. Similarly, the use of feelings, rather than cognition alone can improve an agent’s ability to interpret their environment. When constructing our sentiment indices, we move beyond the dictionary- and lexicon-based methods that have dominated much of the economics and finance literature and instead use transformer-based language models from the modern NLP literature (Vaswani et al. (2017); Devlin, Chang, Lee, and Toutanova (2019)). These models are pre-trained on large corpora and then adapted to the economic and financial domain through fine-tuning and knowledge distillation (Araci (2019); Huang, Wang, and Yang (2023); Sanh, Debut, Chaumond, and Wolf (2019); Gururangan et al. (2020)), which allows them to capture context-dependent meaning that fixed word lists cannot. We apply these methods to a rich corpus of U.S. economic and policy documents in order to construct three complementary narrative sentiment indices, the details of which are described in Section 2.

In order to assess the usefulness of our indices, we evaluate their marginal predictive content by carrying out a series of prediction experiments. In our first set of experiments, we construct real-time macroeconomic and financial predictions for a broad group of target variables, including industrial production, inflation, interest rates, term spreads, equity returns, and credit and volatility measures. These predictions are formulated using models that both contain and omit various sentiment indices, including our three new measures as well as two standard consumer sentiment based measures (the University of Michigan survey of consumer sentiment and the survey based conference board consumer confidence index). This allows us to assess the marginal predictive content of our three new and the two extant indices, when conditioning on other real-time macroeconomic and financial variables. Even if we find that our new indices are found to contain marginal predictive content for econometric models, however, it remains to ascertain whether they contain information that can be used to improve judgement type forecasts. In light of this, in our second set of experiments we evaluate the interplay between judgement and sentiment by examining quasi-judgemental forecasts from the United States Survey of Professional Forecasters (US-SPF). In particular, we carry out forecast rationality tests in order to assess whether US-SPF predictions incorporate the information contained in all five sentiment indices.¹

¹Clements, Rich, and Tracy (2023) surveys the literature on the use of Surveys of Professional Forecasters from the United States and from Europe (ECB-SPF) in the context of using these forecasts to elicit information about expectations and discusses the nature of these forecasts with relation to judgement. Studies by the European Central

The findings of our empirical experiments can be summarized as follows. First, our real-time forecasting experiments indicate that the AR(SIC) benchmark is “MSFE-best” in 11 of 44 target variable / forecast horizon cases that we consider. Among the 33 other “MSFE-best” models, more than half are models containing our new sentiment indices, while most of the other winning models including extant consumer sentiment indices. This is taken as strong evidence that sentiment indices, including both our newly proposed ones and extant consumer survey type indices are useful for real-time econometric forecasting. Second, our rationality tests indicate that there are various instances where SPF forecasts can be improved by using existing real-time macroeconomic information. However, with the exception of a handful of efficiency rejections associated with including consumer sentiment indices in our test regressions, we find a limited number of rejections. This is consistent with SPF forecasts already incorporating much of the information summarized by our narrative sentiment indices. Taken together these findings suggest that our new narrative indices based on NLP and LLM methods contain useful information for both econometric model-based forecasters and professional forecasters.

The rest of the paper is structured as follows. In Section 2, we explain how we use transformer-based language models, LLM-assisted annotation, and zero-shot topic classification to construct our narrative sentiment indices. In Section 3, we describe the real-time data used in our experiments, and discuss the importance of carrying out truly real-time forecasting experiments when comparing judgement-based and model-based forecasts. Section 4 contains details on our empirical setup and Section 5 summarizes our findings based on our empirical analysis. Finally, Section 6 provides concluding remarks. Detailed forecasting and efficiency-test results are collected in the Appendix.

2 Three New Sentiment Indices Based on NLP and LLM Methods

Using textual data to measure sentiment from economic and financial text has a long history. Early measures relied on word lexicons. The Harvard General Inquirer (Harvard GI, subsequently Harvard-IV) project of Stone, Dunphy, Smith, and Ogilvie (1966) constructed a lexicon consisting of ‘positive’ and ‘negative’ words. This list was rather large and represents something of a blunt instrument for measuring the underlying sentiment contained in text. Loughran and McDonald (2011) noted that this list was not well suited to measuring sentiment in financial statements, as words such as ‘liability’ have a specific and not necessarily negative context when reading financial statements, and accordingly pared down the Harvard-IV lexicon into a lexicon better applied to

Bank (de Vincent-Humphreys, Dimitrova, Falck, & Henkel, 2019) and by Stark (2013) describe the mix of formal modeling and judgement in the US-SPF and the European ECB-SPF. In particular both Stark (2013) and an ECB report based on a 2008 survey reports that forecasts contain a mix of econometric forecasts and subjective adjustments. In a 2019 report on the third ECB survey of forecasters, the differences between short-horizon and longer-horizon forecasts are described. For shorter-horizon forecasts, professional forecasters reported that they use a mix of reduced-form econometric models and judgement, while for longer-horizon forecasts, professional forecasters use a mix of structural econometric models and judgment. This paper does not distinguish between the use of judgment to adjust a forecast from an econometric model and the use of judgement to select the appropriate structural model for longer forecast horizons. Instead, we assume that SPF forecasts are “quasi-judgemental”, and they are used to proxy for purely judgemental forecasts in the sequel.

financial text. In the macro-finance setting, lexicon- and dictionary-based methods underlie a number of influential measures, including the economic policy uncertainty index of Baker et al. (2016), the news-sentiment measures of Tetlock (2007) and A. H. Shapiro et al. (2022), and the narrative ‘approach/avoidance’ lexicon of Tuckett, Smith, and Nyman (2014). Our analysis of sentiment connects more broadly to work in behavioral finance studying how sentiment or biased beliefs impact asset prices, as discussed in Barberis (2018), and to studies of the impact of sentiment on the macro-economy, as discussed in Lagerborg, Pappa, and Ravn (2023).

While dictionary methods are transparent and easy to reproduce, they assign a fixed polarity to each word and therefore cannot account for context, negation, or domain-specific usage. We instead construct our sentiment indices using transformer-based language models, which have become the dominant paradigm in natural language processing since the introduction of the attention mechanism by Vaswani et al. (2017) and the bidirectional pre-training approach of Devlin et al. (2019). A key advantage of these models is that the same architecture can be specialized to a particular domain through additional pre-training and fine-tuning (Gururangan et al. (2020)), and can be compressed into smaller, faster models through knowledge distillation (Hinton, Vinyals, and Dean (2015); Sanh et al. (2019)) without a large loss of accuracy. These developments have made it feasible to apply rich, context-aware sentiment classifiers to economic and financial text at scale; see Gentzkow, Kelly, and Taddy (2019) for a survey of text-as-data methods in economics, and Ke et al. (2026) and Bybee et al. (2024) for applications to asset pricing and the business cycle.

Our sentiment indices are constructed from a corpus of U.S. economic and policy documents drawn from public, official sources: Federal Reserve Beige Books and FOMC minutes, Bureau of Labor Statistics Employment Situation and CPI releases, and U.S. Census Bureau retail sales releases. The final corpus contains 1,609 documents and is designed to span major sources of economic narrative that professional forecasters are likely to read. A subset of older Census retail-sales releases are available from the official archive only as image-based PDF scans. For 96 such releases, native PDF text extraction yields no machine-readable text, so we apply OCR to the first-page press-release narrative. The resulting monthly sentiment series used in the final data build span the period 1995:2 - 2024:12. Figure 3 (in the Appendix) displays an overall EconBERT word-cloud diagnostic, and Figures 4-5 provide corresponding topic-specific diagnostics. The overall cloud emphasizes terms associated with EconBERT tone. In the topic-specific clouds, word size emphasizes that how prevalent a term is in highly relevant chunks for a given topic relative to the remainder of the corpus, while color conveys its average EconBERT tone. The topic clouds are therefore intended as descriptive illustrations of the language associated with each economic topic, rather than as fixed lexicons or inputs used directly in constructing the sentiment indices. Our three new indices are described below.

FinBERT (finance-domain) index: The first index, $S_{FB,t}$, is constructed using FinBERT, a finance-domain transformer that adapts the BERT architecture to financial text. FinBERT was introduced by Araci (2019) and further developed by Huang et al. (2023), who pre-train BERT on a large corpus of financial communications and fine-tune it for three-way sentiment classification

(positive, negative, neutral) using labeled financial sentences such as those in the Financial Phrase-Bank of Malo, Sinha, Korhonen, Wallenius, and Takala (2014). For each document in our corpus we split the text into token-bounded excerpts, apply the FinBERT classifier to obtain a probability distribution over the three sentiment classes for each chunk, and define a chunk-level polarity score as the probability of the positive class minus the probability of the negative class. Chunk scores are then averaged within each document and aggregated to the monthly frequency.

EconBERT (economics-tone) index: The second index, $S_{EB,t}$, is designed to capture sentiment in the language of official economic communications rather than market-facing financial news, and is constructed using a large language model (LLM) as an annotator. We first prompt a general-purpose LLM, using a detailed macroeconomic-tone rubric, to label excerpts from our economic-text corpus (Federal Reserve Beige Books and FOMC minutes, Bureau of Labor Statistics releases, and U.S. Census Bureau releases) according to their dominant directional macroeconomic implication: positive (improving activity or easing pressures), negative (deteriorating activity, tightening conditions, or rising stress), or neutral (descriptive, mixed, or directionally ambiguous). Using LLMs in this way as text annotators has been shown to match or exceed human annotators on a range of classification tasks (Gilardi, Alizadeh, and Kubli (2023); Ding et al. (2023)). We then use these LLM-generated labels as supervision to fine-tune a compact transformer classifier, based on DistilBERT (Sanh et al. (2019)), for three-way sentiment classification. This procedure can be viewed as a form of knowledge distillation, in which information from the large LLM teacher is transferred, through generated labels, to a smaller and less computationally costly student model that can be deployed at scale (Hinton et al. (2015)). To preserve a strict out-of-sample design, teacher labels used for model development are restricted to documents available before 2013:01. We use documents available before 2011:01 for the development fit and documents from 2011:1 - 2012:12 for validation; once model choices are fixed, a fresh student model is refit on all eligible labeled excerpts available before 2013:01 and then frozen. Documents dated 2013:01 onward are used for inference only and never update the production classifier. This fixed-model design provides a transparent real-time benchmark, since all out-of-sample observations are evaluated using the same classifier estimated exclusively from training information. Two extensions may be useful for future research. First, one could consider a rolling or expanding-window adaptation scheme in which the student model is periodically updated using only information available up to each forecast origin. Such an approach could accommodate gradual changes in economic language while maintaining a real-time information set, consistent with work on temporal adaptation of transformer models (Röttger and Pierrehumbert (2021)). Second, the LLM-generated training labels could be supplemented with an independently human-annotated sample and the student model trained using both sources of supervision. Hybrid human-LLM annotation schemes may provide a useful way to retain the scalability of automated labeling while anchoring the classifier to human judgments and reducing the influence of systematic errors in teacher-generated labels (Rouzegar and Makrehchi (2024)). Importantly, such human annotations should ideally be collected independently of the LLM-generated labels, since exposure to model suggestions can itself affect the distribution

of human annotations (Schroeder, Roy, and Kabbara (2025)). We leave both extensions to future research. Document-level scores are then formed and aggregated to the monthly frequency. The motivation for a dedicated economics-tone model follows evidence from the computational-linguistics literature on central-bank communication that the tone of official releases contains economically meaningful information (Hansen, McMahon, and Prat (2018); Gorodnichenko, Pham, and Talavera (2023)).²

Topic-Fused index: The third index, $S_{TF,t}$, is a topic-conditioned narrative measure that combines a measure of *what* each excerpt is about with a measure of its *tone*. Rather than discovering topics from the data with unsupervised methods such as latent Dirichlet allocation (Blei, Ng, and Jordan (2003)) or embedding-based clustering (Grootendorst (2022)), we fix a small set of eight pre-specified macroeconomic topics: inflation and prices, the labor market, real activity, monetary policy, financial conditions, housing, consumption and retail sales, and uncertainty and downside risks. These topics are chosen for their direct relevance to our forecast targets. Each chunk is assigned soft topic membership weights using a zero-shot text classifier based on natural-language inference (Yin, Hay, and Roth (2019)), implemented with a pre-trained entailment model (Lewis et al. (2020)); this allows topic assignment without any topic-specific labeled training data. Chunk tone is taken from the distilled EconBERT classifier described above (the positive minus negative probability). For each topic, a topic-specific tone score is formed as the topic membership weighted average of excerpt tone. The monthly Topic-Fused index is then the equal-weight average of the eight topic-specific tone scores. Thus, topic relevance determines how chunks enter each topic-specific tone measure, while the final aggregation gives each of the eight macroeconomic topics equal weight. An alternative direction would be to allow the topic representation itself to be informed by the downstream prediction objective. Supervised topic models explicitly incorporate an associated response variable when estimating latent topics, thereby encouraging the extracted topics to retain information relevant for prediction (Blei and McAuliffe (2007)). Related work in financial economics increasingly combines topic modeling with supervised predictive objectives; for example, Ke et al. (2026) develop a supervised text-mining framework in which textual features and topic information are selected and weighted according to their usefulness for predicting asset returns, while Bybee et al. (2024) show that topic-based news attention contains incremental information for macroeconomic and financial forecasting. A useful extension of our framework would therefore be to investigate whether supervised, forecast-target-aware topic representations can extract additional predictive information beyond our economically motivated pre-specified topic structure. We leave this extension to future research as well.

For FinBERT and EconBERT, the baseline monthly indices use source-balanced aggregation: chunk scores are first averaged within documents, document scores are averaged within each source, and the source-level means are then given equal weight within the month. This prevents months containing unusually many documents or chunks from a particular source from mechanically dom-

²Details of EconBERT sample composition, agreement with teacher-generated labels, confusion matrix, and EconBERT Configuration are available upon the request from the authors.

inating the index. Our Topic-Fused index is constructed as described above from the eight topic-specific EconBERT tone measures.

As a benchmark against which to compare the usefulness of our three new indices, our analysis also incorporates two widely used survey-based measures of consumer sentiment: the University of Michigan Consumer Sentiment Index ($S_{UM,t}$) and the Conference Board Consumer Confidence Index ($S_{CB,t}$). The Michigan index is produced as part of the University of Michigan Surveys of Consumers and is based on responses to five questions concerning: (i) households’ current financial situation relative to one year earlier; (ii) their expected financial situation one year ahead; (iii) expected business conditions in the country as a whole over the next twelve months; (iv) expected business conditions over the next five years; and (v) whether current conditions are favorable for the purchase of major household durable goods. Responses to these questions are aggregated into an index scaled relative to a 1966:Q1 base of 100. The Conference Board Consumer Confidence Index is likewise constructed from five survey questions, but places relatively greater weight on business and labor-market conditions: two questions elicit respondents’ assessments of current business conditions and current job availability, while the remaining three concern expectations six months ahead for business conditions, employment, and family income. The resulting index is reported relative to a 1985 base of 100. Although both indices are designed to summarize consumer attitudes, they capture somewhat different dimensions of sentiment. The Michigan index places relatively greater emphasis on household finances, buying conditions, and longer-horizon economic expectations, whereas the Conference Board index contains more direct information about current and expected labor-market conditions. See Bram and Ludvigson (1998) for a detailed comparison of the two surveys and their predictive content, and Ludvigson (2004) for a broader discussion of the measurement and economic information content of consumer confidence indices. The full set of sentiment indicators, together with the macroeconomic and financial target variables, is summarized in Table 1. Figures 1 and 2 plot the sentiment indices alongside the eleven forecasting targets over the out-of-sample period.

3 Data

In our forecasting experiments, we utilize real-time US macroeconomic data contained in the monthly FRED-MD database of McCracken and Ng (2016), which is maintained by the Federal Reserve Bank of St. Louis and contains roughly 130 macroeconomic variables. The full description and transformation applied to each variable are reported in Table 1. In our analysis, we forecast eleven macroeconomic and financial variables including: CPI inflation (cpi_t), the growth rate of industrial production (ip_t), the civilian unemployment rate (UR_t), the 3-month Treasury bill rate ($IR3M_t$), the 10-year Treasury bond rate ($IR10Y_t$), two term spreads ($TS1Y_t$ and $TS10Y_t$), the return on the S&P 500 composite index ($sp500_t$), the CBOE volatility index (vix_t), and both the GZ credit spread (GZ_t) and excess bond premium (EBP_t) of Gilchrist and Zakrajšek (2012). Our historical dataset spans the period 2003:1 - 2024:1. In addition to the full out-of-sample period,

we split the sample into a pre-COVID subsample (2013:1 - 2019:12) and a post-2019 subsample (2020:1 - 2024:1).

Note is that the FRED-MD dataset is truly real-time, in the sense that a new “vintage” of data is available at each calendar date. More specifically, for a given variable, a vintage contains the most recent revision for all of the calendar dated observations in the vector valued dataset. In the vintage, an older observation may have been revised many times, while more recent observations will have been revised fewer times. An example is the January 2020 vintage of unemployment. Assume that the vintage has data beginning in 1990. Then, the vintage contains the most recently available (as of January 2020) datum for January 1990 which may have been revised many times, the most recently available (as of January 2020) datum for February 1990 which may have been revised many times (albeit at most one less time than the January 1990 datum has been revised), and so on until the December 2019 observation, which as of the January 2020 date of the vintage, is the first available (or first release) datum for calendar date December 2019. Evidently, a vintage of data has older observations that have been potentially revised many times and newer dated observations which have been revised significantly fewer times. In our application, the “latest” vintage of data is used prior to the construction of each new forecast of the 11 target variables given above. This simulates a truly real-time procedure where the latest available revision for each observation in a dataset is used in forecast model estimation.

To illustrate further, consider the value of unemployment, this for for the period June 2020. In July 2020, the government reported a “first release” value for June. In August 2020, they updated their “first release” of the June unemployment figure. This new value for June unemployment is called the “second release”. This process of revision continues indefinitely. New releases of older calendar dated observations are regularly published for reasons such as changes in variable definitions (e.g. changes to the basket of goods appearing in the CPI) and newly collected historical data (e.g. surveys of firms that have arrived “late”). As discussed above, a real-time data vintage is a snapshot of a dataset showing exactly what data were available to researchers or policymakers at a particular point in time. Some data, such as financial variables including stock prices and interest rates are not subject to revision. Other variables, such as unemployment, industrial production, and the CPI, are subject to revision. Using real-time data allows us to truly simulate a forecasting environment in which models are updated at each point in time using data that were actually available at that time. Were we to simply collect unemployment data from a website today, calendar dated observations in our dataset from 2020 would reflect revisions that occurred after 2020. For variables that are revised regularly, this means that forecasting experiments of the type carried out in this paper would be invalid, in the sense that they would be utilizing “future data” when estimating forecasting model regressions. This is the reason why we need to use real-time data in our analysis, and leads to the need for notation when denoting our real-time time series variables, which is more general than simply denoting a calendar dated time series observation as y_t , say. The notation will be discussed in the next section. For further discussion of the structure of real-time datasets, as well as methods for real-time forecasting, see Swanson (1996), Swanson, Ghysels, and

Callan (1999), and Kim and Swanson (2018).

As discussed above, in addition to the reporting on prediction experiments using the monthly FRED-MD data discussed above, we carry out forecast rationality tests using quarterly predictions from the U.S. Survey of Professional Forecasters (SPF) published by the Federal Reserve Bank of Philadelphia. The reason for the change from a monthly frequency (when performing real-time econometric model-based forecasting experiments) to a quarterly frequency (when performing rationality tests) is that the SPF reports predictions of experts in their survey and associated historical real-time data at a quarterly frequency. Our historical quarterly SPF dataset spans the period 2011Q1 - 2024Q1. The five target variables used in our SPF forecast efficiency tests are: the growth rate of industrial production (ip_t), CPI inflation (cpi_t), the 3-month Treasury bill rate ($IR3M_t$), the 10-year Treasury bond rate ($IR10Y_t$), and the unemployment rate (UR_t). These variables are a subset of the 11 variables analyzed in our monthly real-time forecasting experiment, and represent all of the overlapping variables when comparing the two datasets. Following numerous papers that examine US-SPF data, we utilize median responses across SPF respondents when carrying out our forecast rationality tests. For further discussion, see Baghestani (2005), Branch and Evans (2006), and Schorfheide and Song (2024). Table 2 summarizes these quarterly variables and transformations.

4 Empirical Methods

In this section, we discuss the experimental setup used in our analysis of: (i) the marginal predictive content of our new sentiment variables in a real-time forecasting experiment; and (ii) the efficiency of quasi-judgement based predictions made by respondents to the Survey of Professional Forecasters, when including sentiment indices and observable macroeconomic variables in real-time forecast efficiency-test regressions.

4.1 Experimental Setup

As discussed above, we utilize monthly real-time variables for econometric model-based estimation and quarterly real-time variables for carrying out rationality tests of SPF survey based forecasts.

When constructing econometric model-based forecasts, the number of lags as well as all model parameters are re-estimated prior to the construction of each new prediction, using rolling 120-month windows of data and models constructed using various sentiment indices and macroeconomic variables. Forecasts are constructed for horizons of $h = 1, 3, 6$, and 12 months ahead. Our full out-of-sample forecasting period is 2013:1 - 2024:1. To assess the stability of our findings, we also report results separately for a pre-COVID subsample (2013:1 - 2019:12) and a post-2019 subsample (2020:1 - 2024:1).

For efficiency testing, our quarterly data span the period 2011Q1 - 2024Q1. For associated test regressions, we use SPF forecasts as well as realizations on our sentiment indices and various macroeconomic variables. Models are not re-estimated at each point in time, as our SPF forecasts

are obtained directly from the Federal Reserve Bank of Philadelphia. Forecasts in these regressions are for horizons of $h = 1, 2, 3$, and 4 quarters ahead. See below for further details.

Recall from Section 2 that our sentiment indices are constructed using transformer-based language models, LLM-assisted annotation, and zero-shot topic classification applied to a large corpus of U.S. economic and policy documents. We group the five sentiment measures into two blocks: a block of three LLM-based narrative indices, $S_{1,t} = (S_{FB,t}, S_{EB,t}, S_{TF,t})'$, and a block of two survey-based consumer-sentiment indices, $S_{2,t} = (S_{UM,t}, S_{CB,t})'$, where $S_{UM,t}$ is the University of Michigan consumer sentiment survey index and $S_{CB,t}$ is the Conference Board consumer sentiment survey index. Additionally, we define $S_t = (S'_{1,t}, S'_{2,t})'$, as detailed in Table 1. These indices are used to investigate whether the information in our new sentiment indices as well as our two extant consumer sentiment indices can be used to improve forecast accuracy (see Section 4.2) and whether quasi-judgmental forecasts are efficient relative to the information contained in S_t (see Section 4.3).

4.2 Econometric Model-based Forecasting

Since we utilize real-time data in our experiments, we require notation that accounts for the release date of a calendar dated observation. Thus, for real-time data that are subject to revision, let $x_t^{(k)}$ denote the release k value of variable x for calendar date t . In this sense, setting $k = 1$ defines preliminary or first release data, while $k = 2$ corresponds to second release data, and so on, for $k = 1, \dots, f$, where f denotes data that are fully revised. Note that this notation only pertains to our macroeconomic data as they are subject to revision. Our sentiment indices are not subject to revision, and hence we simply use our earlier conventional notion for variables such as S_t .

Turning to our forecasting experiment, assume that forecast models used to predict month t are estimated using the real-time data vintage available at time t . Consider a time series y . In period t , the econometrician observes the first release of y_{t-1} ($y_{t-1}^{(1)}$), the second release of y_{t-1} ($y_{t-1}^{(2)}$), the third release of y_{t-1} ($y_{t-1}^{(3)}$) and so on. The general model that is estimated is presented in (1). All of our forecasting models are nested within the following specification:

$$y_{t+h}^{(1)} = \alpha + \sum_{i=1}^{p_0} \phi_i y_{t+1-i}^{(i)} + \sum_{i=1}^{p_1} \beta'_i \mathbf{Y}_{t+1-i}^{(i)} + \sum_{i=1}^{p_2} \delta'_i \mathbf{S}_{1,t+1-i} + \sum_{i=1}^{p_3} \theta'_i \mathbf{S}_{2,t+1-i} + \varepsilon_{t+h}, \quad (1)$$

where $y_{t+h}^{(1)}$ is the scalar target variable being forecasted and $\mathbf{Y}_{t+1-i}^{(i)} = (ip_{t+1-i}^{(i)}, cpi_{t+1-i}^{(i)}, IR3M_{t+1-i}, IR10Y_{t+1-i}, UR_{t+1-i}^{(i)})'$.³ The vector $\mathbf{Y}_{t+1-i}^{(i)}$ contains the growth rate of industrial production, CPI inflation, the 3-month Treasury bill rate, the 10-year Treasury bond rate, and the unemployment rate, respectively. The target variables analyzed in our experiment include CPI inflation ($cpi_{t+h}^{(1)}$), the growth rate of industrial production ($ip_{t+h}^{(1)}$), the civilian unemployment rate ($UR_{t+h}^{(1)}$), the 3-month Treasury bill rate ($IR3M_t$), the 10-year Treasury bond rate ($IR10Y_t$), the 10 year bond and 3 month-Treasury bill spread ($TS1Y_t$), the 1 year bond and 3 month-Treasury bill spread ($TS10Y_t$), the S&P500 returns ($sp500_t$), the CBOE volatility index (vx_t), the GZ credit spread (GZ_t), and

³Note that the maintained assumption is that we are forecasting the first release of a variable. Additionally, our $h = 1$ forecast corresponds to what is often referred to as a “nowcast” in the economics forecasting literature.

an excess bond premium (EBP_t).⁴ Note that the maintained assumption is that in our real-time forecasting experiment we are interested in forecasting the first release ($y_{t+h}^{(1)}$) of a future value of the target variable. When evaluating forecasts, we use forecast errors defined as:

$$e_{t+h} = y_{t+h}^{(1)} - \hat{y}_{t+h}^{(1)}, \quad (2)$$

where $\hat{y}_{t+h}^{(1)}$ denotes the real time forecast of $y_{t+h}^{(1)}$ obtained after estimating equation (1) .

In summary, we estimate eleven different forecasting models, in addition to an AR(SIC) benchmark (see Table 3 for details). Our benchmark AR(SIC) model sets $\beta_i = \delta_i = \theta_i = 0$ for all i in equation (1), retaining only autoregressive terms, with lag chosen using the Schwarz (Bayesian) Information Criterion (SIC). The remaining models augment this benchmark as follows. Models 1-3 add a single LLM-based index ($S_{FB,t}$, $S_{EB,t}$, or $S_{TF,t}$, respectively), and Model 4 adds the full LLM sentiment index block, $S_{1,t}$. Models 5-6 add a single consumer survey-based index ($S_{UM,t}$ or $S_{CB,t}$), and Model 7 adds the full consumer sentiment index block $S_{2,t}$. Model 8 adds the macroeconomic block $\mathbf{Y}_t$ alone. Finally, Models 9-11 combine the macroeconomic block with the LLM block ($S_{1,t}$), the survey block ($S_{2,t}$), and both sentiment blocks ($S_{1,t}$ and $S_{2,t}$), respectively.

To assess whether differences in mean square forecast error (MSFE) are statistically significant, we use the unconditional predictive-ability test of Giacomini and White (2006). Recall that the null hypothesis for this test is the same as that of the Diebold-Mariano test (see Diebold and Mariano (1995)), which is $H_0 : E[l(e_{t+h,i})] - E[l(e_{t+h,j})] = 0$, where $e_{t+h,i}$ is our benchmark AR model and $e_{t+h,j}$ is a competitor model. In our analysis, $l(\cdot)$ is a quadratic loss function and the test statistic is:

$$GW_P = \sqrt{P} \frac{\bar{d}}{\hat{\sigma}_d}, \quad (3)$$

where $\bar{d}$ denotes the mean of d_{t+h} , and $d_{t+h} = (e_{t+h,i})^2 - (e_{t+h,j})^2$. Here, $\hat{\sigma}_d$ is a heteroskedasticity and autocorrelation consistent estimate of the standard deviation of d_{t+h} , and P denotes the number of out-of-sample predictions used to construct the test statistic. A significantly positive value of GW_P indicates that the competing model is preferred to the AR(SIC) benchmark. In the sequel, we assume the statistic is asymptotically standard normal following Giacomini and White (2006). See also Clark and McCracken (2001), Rossi and Sekhposyan (2011) and Corradi and Swanson (2013) for further discussion of forecast accuracy comparison testing. In addition, we construct 90% model confidence sets (MCS) following Hansen, Lunde, and Nason (2011). The MCS procedure starts with all models and repeatedly tests the null hypothesis that the remaining models have equal predictive ability. If the null hypothesis is rejected, the worst-performing model is removed. The process continues until the equal-predictive-ability null can no longer be rejected.

⁴When our “real-time” notation is not used, a variable is not subject to revision.

4.3 Efficiency Testing Using Judgement Based Forecasts

As discussed above and in Clements et al. (2023), professional forecasters including those surveyed in the Survey of Professional Forecasters maintained by the Federal Reserve Bank of Philadelphia provide point predictions that are informed by both models and judgement for specific core macroeconomic variables at various forecast horizons. The same is true for the ECB-SPF. For example, Clements et al. (2023) point out that surveys conducted by the ECB in 2008, 2013, and 2018 indicated a prevalence of judgement based forecasters among survey participants, although the number of model-based forecasters appeared to be increasing over time (see de Vincent-Humphreys et al. (2019) for further discussion). For these reasons, a natural vehicle for assessing the predictive linkage between judgment and sentiment is the SPF, with the caveat that forecasts in this survey are not solely judgement-based. It is for this reason that we often refer to SPF forecasts as quasi-judgemental.

In order to provide initial evidence of the usefulness of our new sentiment indices for judgemental forecasting, we thus utilize SPF forecasts and draw on the literature on forecast rationality testing. In particular, we carry out so-called efficiency test regressions (see e.g. Clements (1995), Swanson and Van Dijk (2006) and the references cited therein for further discussion).

More specifically, we test the rationality of SPF forecasts, and by proxy the rationality of judgment forecasts, conditional on sentiment (and the period t vintage of the macroeconomic conditioning variables in $\mathbf{Y}$). Early papers in the efficiency testing literature followed (Nordhaus, 1987), and fitted regressions of the form:

$$y_{t+h} = \alpha + \beta \hat{y}_{t+h} + \epsilon_{t+h}, \quad (4)$$

where $\hat{y}_{t+h}$ is a horizon h forecast and y_{t+h} denotes the actual observation on some target variable, y . When evaluating whether forecasts were unbiased and efficient (i.e., whether $\alpha = 0$ and $\beta = 1$, respectively) heteroscedasticity and autocorrelation consistent (HAC) standard errors were generally used for multi-step ahead forecasts, $h > 1$. In this case, “rationality” was deemed “true” there was no bias and no inefficiency. This literature has seen many contributions over the last few decades. A key paper, for example, is Romer and Romer (2000) who propose testing forecast efficiency using real-time variables. In Swanson and Van Dijk (2006), the authors propose a forecast efficiency test that allows the investigator to determine the “timing” of data rationality, in the sense that one can assess how many months it takes before an early release of a variable is efficient, in the sense that all information available at the time of the first release of the data has been properly conditioned on. Faust and Wright (2008) construct two versions of OLS-based tests called “naive” and “transparency”, and use conventional Wald statistics to test for forecast efficiency. In Ellingsen, Larsen, and Thorsrud (2022), the authors show that SPF forecasts under-perform traditional (real-time) econometric based forecasts for several key macroeconomic variables, while Corradi and Swanson (2013) introduce distribution based tests based on stochastic dominance principles for assessing the accuracy of SPF predictions.

In our experiment, we evaluate the (bias and) efficiency of SPF forecasts. We condition not only on macroeconomic variables available at the time forecasts were made, but also on our sentiment indices. This approach allows us not only to re-assess the efficiency of SPF forecasts using a standard macroeconomic dataset, but also using our new sentiment indices. Namely, we are able to assess whether efficiency with respect to macroeconomic variables carries over to sentiment indices: is a forecast that is efficient when accounting for real-time macroeconomic information still efficient when sentiment is added to the conditioning information set? Our test regression in this context is a variant of that discussed above, and follows most closely that used in Swanson and Van Dijk (2006). See also Genre, Kenny, Meyler, and Timmermann (2013) for further discussion of SPF forecasting evaluation methodology.

Let t denote the reference quarter for the SPF survey. Forecasts $\hat{y}_{t+h}^{SPF}$ are made with information up to and including information available at the beginning of period $t + 1$. This mimics our above framework, as $h = 1$ forecasts are so-called “nowcasts”. Let $e_{t+h}^{SPF} = y_{t+h}^{(1)} - \hat{y}_{t+h}^{SPF}$ denote the SPF forecast error of the target for quarter $t + h$, where $y_{t+h}^{(1)}$ denotes the realization of the target variable for quarter $t + h$. Thus, $h = 1, 2, 3, 4$ denote one- to four-quarter-ahead forecasts relative to the reference quarter t . Our test regression takes the form:

$$e_{t+h}^{SPF} = \alpha + \beta' S_t + \gamma' \mathbf{Y}_t^{(1)} + \varepsilon_{t+h}, \quad (5)$$

where $\mathbf{Y}_t^{(1)}$ is as defined above. Here, $\mathbf{Y}_t$ collects, for each macroeconomic variable, the latest monthly observations through reference quarter t , using the first-month vintage of the corresponding SPF survey quarter, or its quarterly vintage for unemployment. The sentiment variable S_t denotes the sentiment block aligned with reference quarter t , either the LLM block $S_{1,t}$ or the survey block $S_{2,t}$.

Recall also that while our above forecasting experiments utilize monthly data, our efficiency test analysis is based on the use of quarterly data, given that SPF forecast surveys are carried out on a quarterly basis. For the quarterly macroeconomic variables, we adopt a simple real-time timing convention, the information set at SPF survey quarter is represented by the first calendar-month vintage of that quarter, except for unemployment, for which we use the corresponding quarterly vintage. For narrative sentiment, the availability-based alignment described in Section 3 implies that the January, April, July, and October monthly observations correspond to Q1-Q4 SPF survey origins, respectively. These observations are indexed by the preceding reference quarter t in equation (5). We estimate two versions of equation (5): one in which S_t contains the three LLM-based narrative indices, $S_{1,t}$, and one in which S_t contains the two survey-based indices, $S_{2,t}$.

The forecast efficiency tests that we carry out are robust Wald tests of the following hypotheses: $H_0 : \alpha = 0, \beta = \mathbf{0}, \gamma = \mathbf{0}$; $H_0 : \alpha = 0$; $H_0 : \beta = \mathbf{0}$; and $H_0 : \gamma = \mathbf{0}$. Note that if $\alpha \neq 0$ then forecasts are biased, while rejections due to $\beta \neq \mathbf{0}$ or $\gamma \neq \mathbf{0}$ imply inefficiency, in the sense that variables available at the time forecasts were made contained information that could have been used by forecasters to improve the accuracy of their predictions. Finally, in addition to reporting results based on all of these different tests, we separately test efficiency using each of our three new

LLM sentiment indices as well as using each of the two consumer sentiment indices.

5 Empirical Findings

5.1 Results From the Real-Time Forecasting Experiment

Our forecasting experiments were performed using the eleven models (and the AR(SIC) benchmark) described in Section 4.2, with a rolling window of 120 monthly observations used to re-estimate each model prior to the construction of each new monthly forecast. The initial estimation window is January 2003 - December 2012. The forecasting model specifications are reported in Table 3. The number of lags is chosen anew using the SIC for each estimation window. Forecasts are constructed for 1, 3, 6, and 12 months ahead.

Table 4 reports results for the full out-of-sample period (2013:1 - 2024:1). In this table, MSFEs relative to the AR(SIC) benchmark are reported. Thus, a relative MSFE less than 1 for a given model is evidence that the model outperforms the benchmark. Also reported in Table 4 are Giacomini and White (GW) test results for the hypothesis that a given model's MSFE is significantly different from that of the benchmark. Inspection of the results in this table indicate that the AR(SIC) benchmark outperforms our sentiment-augmented specifications in only 11 of the 44 target variable / forecast horizon cases (recall that there are 4 forecast horizons and 11 target variables, and thus 44 target variable / forecast horizon permutations), while at least one sentiment augmented specification improves upon the benchmark in the remaining 33 cases. Among these 33 non-AR(SIC) wins, 15 are obtained by Models 1-4, which augment the benchmark with LLM-based sentiment indices only; 10 are obtained by Models 5-7, which augment the benchmark with survey-based consumer sentiment indices only; and 2 are obtained by Model 8, which includes additional macroeconomic regressors but no sentiment indices. The remaining 6 wins are obtained by the models that combine the macroeconomic variables with sentiment: Model 9 (macroeconomic variables and the LLM-based sentiment block), Model 10 (macroeconomic variables and the survey-based sentiment block), and Model 11 (macroeconomic variables and both sentiment blocks) each win in 2 cases. Thus, of the 33 cases in which a non-AR(SIC) model has the lowest MSFE, 31 involve a specification containing at least one sentiment index, while 2 correspond to the macroeconomic variable only specification. Classifying the winning models according to the type of sentiment index that they contain, LLM-based indices appear in 19 of the 33 winning specifications (Models 1-4, 9, and 11) and survey-based indices appear in 14 (Models 5-7, 10, and 11), where the 2 wins of Model 11 are included in both totals. Taken together, this evidence points to the clear usefulness of both our new sentiment indices and our widely used survey based sentiment indices.

Tables 5 and 6 report corresponding results for pre-COVID (2013:1 - 2019:12) and post-2019 (2020:1 - 2024:1) subsamples. Comparing these results with the full-sample results in Table 4, the AR(SIC) benchmark has the lowest MSFE considerably more often in the pre-COVID period than in the post-2019 period. Specifically, the AR(SIC) benchmark wins in 19 of the 44 cases in the pre-COVID sample, with one of the eleven alternative models winning in the remaining 25 cases.

Among these 25 non-AR(SIC) wins, Models 1-4 account for 15 wins and Models 5-7 account for 7 wins. Model 8, the only specification with solely macroeconomic variables, does not win in any case. Of the models combining macroeconomic variables with sentiment indices, Model 10 wins in 2 cases and Model 11 wins in 1 case, while Model 9 does not win in any case. Thus, all 25 non-AR(SIC) wins in the pre-COVID sample are obtained by specifications containing at least one sentiment index, with LLM-based indices appearing in 16 of the winning specifications and survey-based indices appearing in 10. In the post-2019 sample, by contrast, the AR(SIC) benchmark has the lowest MSFE in only 3 of the 44 cases, while one of the eleven alternative models wins in the remaining 41 cases. Among these 41 non-AR(SIC) wins, Models 1-4 account for 17 wins and Models 5-7 account for 12 wins. The macroeconomic variable only specification, Model 8, wins in 6 cases. Finally, Models 9, 10, and 11 account for 2, 3, and 1 wins, respectively. Thus, sentiment-containing specifications account for 35 of the 41 non-AR(SIC) wins in the post-2019 sample, with LLM-based indices appearing in 20 of the winning specifications and survey-based indices appearing in 16, while the remaining 6 wins are attributable to the macroeconomic-variable-only specification. In this sense, the importance of sentiment indices such as those introduced in this paper appears to be increased in recent years.

Of final note is that our Model Confidence Set results complement the above findings, showing that sentiment-augmented specifications remain competitive (i.e. are in the final confidence sets) in all 44 cases. Complete MCS results are available upon request from the authors.

To summarize, there is broad evidence that sentiment information beyond that contained in the AR(SIC) benchmark improves forecast performance in econometric forecasting models. Across the full sample, the AR(SIC) benchmark has the lowest MSFE in 11 of the 44 target/horizon cases, while one of the eleven alternative models has the lowest MSFE in 33 cases. Of these 33 cases, 31 are associated with models containing at least one sentiment index and 2 with the macroeconomic variable only model. Subsample results show that the relative performance of the AR(SIC) benchmark differs substantially across periods: it wins in 19 cases before COVID but in only 3 cases after 2019. Sentiment-containing specifications remain important in both subsamples, while the macroeconomic variable only specification becomes considerably more competitive in the post-2019 period.

5.2 Evaluating SPF Forecasts

We next evaluate the rationality of the quasi-judgemental forecasts from the Survey of Professional Forecasters, when information from both macroeconomic variables as well as sentiment indices is used to assess forecast efficiency. This is important given our prior findings. Namely, we now know that our new sentiment indices contain marginal predictive content when utilized in econometric forecasting models. However, we do not know whether the information contained in the indices is utilized by professional forecasters. If it is not, then professional forecasters could also benefit from using the new indices. If it is, however, then there is evidence that professional forecasters have already been utilizing the information contained in the indices. In our experiments, we include

both sentiment indices and real-time macroeconomic conditioning variables, analogous to our earlier real-time forecasting experiments (see Section 4.3 for a discussion of the regressions reported on in this section).

Summary results are reported in Table 7 for our three LLM-based narrative indices and in Table 8 for the two survey-based sentiment indices. For each target variable and forecast horizon, the tables report rejections and associated p-values for the four hypotheses discussed in Section 4.3: “overall” rationality (i.e. unbiasedness + efficiency) - $H_0 : \alpha = \beta = \gamma = 0$; unbiasedness - $H_0 : \alpha = 0$; and efficiency - $H_0 : \beta = 0$ and $H_0 : \gamma = 0$. Robust t-statistics and Wald F-tests are used for the tests, and all rejections are evaluated at the 10% level. Finally, note that SPF forecasts are quarterly, the forecast horizons range from $h = 1$ to 4 quarters ahead, and the full sample period is 2011Q1–2024Q1. Note that results for various subsamples are reported in a supplemental appendix.

Turning to our findings, first consider regressions using our three LLM-based narrative sentiment indices, as reported in Table 7. The “overall” rationality hypothesis is rejected in 14 of the 20 target variable / forecast-horizon cases. For $h = 1$, the null hypothesis is rejected for three of the five target variables: CPI inflation, the 10-year Treasury bond rate, and unemployment. The same three rejections occur for $h = 2$. At $h = 3$ and $h = 4$, the 3-month Treasury bill rate joins this group, resulting in four rejections at each of these horizons. Industrial production is the only target variable for which the “overall” rationality hypothesis is not rejected at any horizon. Importantly, the 14 “overall” rejections do not imply that the LLM-based sentiment indices themselves contain omitted information in 14 cases. Some rejections may be due to bias and others may be due to omitted macroeconomic variables. When we test for efficiency using only the LLM-based sentiment indices (i.e. $H_0 : \beta = 0$), the null is rejected in only 4 of 20 regressions: for unemployment at $h = 1$ and $h = 4$, and for the 3-month Treasury bill rate at $h = 2$ and $h = 3$. In contrast, using only the macroeconomic variables (i.e. $H_0 : \gamma = 0$), the null is rejected in 11 of 20 regressions. Additionally, unbiasedness (i.e. $H_0 : \alpha = 0$) is rejected in only 1 of 20 cases, for unemployment at $h = 1$. Thus, although the overall rationality hypothesis is rejected relatively frequently, our evidence suggests that these rejections are driven primarily by information contained in the real-time macroeconomic variables rather than by omitted information in the LLM-based narrative sentiment indices. These results are consistent with SPF forecasts already incorporating much of the information summarized by our narrative sentiment indices. Just as in the case of our real-time forecasting experiments, this result underscores the potential importance of the information captured in our new sentiment indices.

Table 8 reports corresponding results for the case where our other two sentiment indices, the University of Michigan Consumer Sentiment Index and Conference Board Consumer Confidence Index, are used in place of our LLM-based narrative indices. The overall rationality hypothesis is rejected in 12 of the 20 target variable / forecast horizon cases. At $h = 1$, the null is rejected for CPI inflation, the 10-year Treasury bond rate, and unemployment. At $h = 2$ and $h = 3$, it is rejected for the 3-month Treasury bill rate, the 10-year Treasury bond rate, and unemployment.

At $h = 4$, it is rejected for CPI inflation and both interest-rate variables. As in the LLM-based specification, the overall rationality hypothesis is never rejected for industrial production. This suggests that although these two sentiment indices are already publicly available, SPF forecasters would benefit by utilizing the information contained in them.

Taken together, the full-sample results in Tables 7 and 8 reveal a clear distinction between the two types of sentiment information. The overall rationality hypothesis is rejected at similar frequencies in the two specifications, 14 of 20 cases, with LLM-based sentiment and 12 of 20 cases with survey-based sentiment. However, the corresponding sentiment-only tests differ markedly. LLM narrative sentiment index based efficiency is rejected in only 4 of 20 cases, whereas the survey-based sentiment index based efficiency is rejected in 12 of 20 cases. As discussed above, this suggests that SPF forecasts incorporate much of the information contained in our contemporaneous economic narrative measures, while the University of Michigan and Conference Board indices retain incremental information for a broader set of SPF forecast targets. One possible explanation is that professional forecasters observe much of the official-release and policy-related information from which our narrative indices are constructed and may therefore already incorporate this information when forming their forecasts. By contrast, the consumer-sentiment surveys may summarize dimensions of household expectations and confidence that are less completely reflected in professional forecasts. Thus, our recommendation based on these findings is that LLM type sentiment indices contain information that is useful, hence warranting their regular publication, and that “already-published” consumer sentiment indices could be better utilized by SPF forecasters.

To examine whether these findings are stable across different sample periods, Supplemental Appendix Tables 1–10 repeat the above analysis separately for a pre-COVID subsample (2011Q1–2019Q4) and a post-2019 subsample (2020Q1–2024Q1). The pre-COVID results are broadly consistent with the conclusion that LLM narrative type sentiment contributed relatively little incremental information to SPF forecast errors during this period. For the LLM-sentiment augmented specifications, the overall rationality hypothesis is rejected in 13 of 20 cases, compared with 14 of 20 cases in the full sample. However, LLM-sentiment based rejections occur in only 3 of 20 cases, macroeconomic variable based rejections occur in 10 of 20 cases, and the unbiasedness hypothesis is rejected in only 1 case. For consumer-sentiment augmented specifications, the overall rationality hypothesis is rejected in 11 of 20 cases, but consumer sentiment based rejections do not occur in any of the 20 regressions. Macroeconomic variables are associated with rejections in 9 cases. Thus, particularly for the survey-based measures, pre-COVID evidence indicates that the overall rationality rejections are associated primarily with the macroeconomic conditioning information rather than with omitted sentiment information.

The results change markedly in the post-2019 subsample. For the LLM-sentiment augmented specifications, the overall rationality hypothesis is rejected in all 20 target variable / forecast horizon cases, compared with 13 of 20 before 2020 and 14 of 20 in the full sample. LLM-sentiment based rejections occur in 10 of 20 cases, macroeconomic variable based rejections in 18 of 20 cases, and the unbiasedness hypothesis is rejected in 4 cases. For consumer-sentiment augmented specifications,

the overall rationality hypothesis is rejected in 18 of 20 cases, compared with 11 of 20 cases in the pre-COVID sample. Consumer sentiment based rejections occur in 10 of 20 cases, macroeconomic variable based rejections in 18 of 20 cases, and the unbiasedness hypothesis is rejected in 12 cases. Hence, the evidence of forecast irrationality and in particular forecast inefficiency becomes substantially stronger after 2019 for both types of sentiment indices. The change is particularly notable for the consumer-sentiment augmented specifications, where the number of rejections increases from zero in the pre-COVID sample to 10 of 20 in the post-2019 sample. These post-2019 results should nevertheless be interpreted with caution. The subsample contains only 17 quarterly observations, and HAC-based Wald tests may have poor finite-sample properties in samples of this size. Accordingly, we place greater weight on our full-sample evidence, while viewing the subsample results as indicative of possible changes in the information content of sentiment surrounding the COVID and post-COVID periods.

As a further robustness check, we repeated the entire analysis using the Philadelphia Fed’s official realization benchmark. Namely, we used real-time data downloaded from the Philadelphia Federal Reserve Bank in order to obtain “actual realizations” of the variables being predicted by the SPF forecasters. In this robustness check, our “actual realizations” were replaced with data taken directly from the Survey of Professional Forecasters. While the quantitative results differed marginally from those reported in this paper, our qualitative findings remain exactly the same and do our conclusions.⁵

6 Discussion and Conclusion

In this paper we construct three new narrative sentiment indices based on the use of NLP and LLM methods. We analyze these indices, alongside widely used survey-based measures of consumer sentiment, to investigate whether the information contained in sentiment can improve econometric model-based forecasts and whether quasi-judgemental forecasts are efficient relative to the information contained in such sentiment indices. Our results suggest that both our new sentiment indices as well as extant consumer-based sentiment indices contain marginal predictive content that can be used to improve econometric model-based forecasts. However, the scope for improving quasi-judgemental SPF forecasts depends on the source of sentiment. Much of the information contained in our NLP and LLM-based sentiment indices appears to already be incorporated in SPF forecasts, whereas consumer-based sentiment indices sometimes retain incremental information that could be used to further improve them. In summary, the usefulness of our new indices is supported by both our real-time econometric model-based forecasting findings and by the fact that our efficiency test findings are consistent with SPF forecasters already incorporating much of the information summarized by our narrative sentiment indices.

This research is meant as a starting point, as many questions remain to be answered. For example, a natural direction for future work is to explore richer LLM architectures and prompting

⁵Full results of this robustness analysis are available upon request from the authors.

strategies, extend our topic-conditioned approach to a finer set of economic themes, and study how the marginal value of text-based sentiment evolves with the information environment. Additionally, it remains to more deeply explore the interplay between NLP and LLM type sentiment indices and various survey based sentiment indices. For example, it would be of interest to explore ways to combine real-time survey based information into our NLP and LLM type indices.

References

- Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.
- Baghestani, H. (2005). Improving the accuracy of recent survey forecasts of the t-bill rate: other survey information can help. *Business Economics*, 40, 36–40.
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131(4), 1593–1636.
- Barberis, N. C. (2018). Psychology-based models of asset prices and trading volume. In B. D. Bernheim, S. DellaVigna, & D. Laibson (Eds.), *Handbook of Behavioral Economics: Foundations and Applications 1* (pp. 79–176).
- Blei, D. M., & McAuliffe, J. D. (2007). Supervised topic models. In *Advances in Neural Information Processing Systems* (Vol. 20, pp. 121–128).
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Bram, J., & Ludvigson, S. C. (1998). Does consumer confidence forecast household expenditure? a sentiment index horse race. *Federal Reserve Bank of New York Economic Policy Review*, 4(2), 59–78.
- Branch, W. A., & Evans, G. W. (2006). A simple recursive forecasting model. *Economics Letters*, 91(2), 158–166.
- Bybee, L., Kelly, B. T., Manela, A., & Xiu, D. (2024). Business news and business cycles. *The Journal of Finance*, 79(5), 3105–3147.
- Clark, T. E., & McCracken, M. (2001). Tests of equal forecast accuracy and encompassing for nested models. *Journal of Econometrics*, 105(1), 85–110.
- Clements, M. P. (1995). Rationality and the role of judgment in macroeconomic forecasting. *Economic Journal*, 105, 410–420.
- Clements, M. P., Rich, R. W., & Tracy, J. S. (2023). Surveys of professionals. In R. Bachmann, G. Topa, & W. van der Klaauw (Eds.), *Handbook of Economic Expectations* (pp. 71–106). Elsevier.
- Corradi, V., & Swanson, N. R. (2013). A survey of recent advances in forecast accuracy comparison testing, with an extension to stochastic dominance. In *Recent Advances and Future Directions in Causality, Prediction, and Specification Analysis: Essays in Honor of Halbert L. White jr* (pp. 121–144). Springer.
- de Vincent-Humphreys, R., Dimitrova, I., Falck, E., & Henkel, L. (2019). Twenty years of the ECB survey of professional forecasters. *ECB Economic Bulletin*, 1, 34–58.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: Human language technologies, volume 1* (pp. 4171–4186). Association for Computational Linguistics.

- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Ding, B., Qin, C., Liu, L., Chia, Y. K., Li, B., Joty, S., & Bing, L. (2023). Is GPT-3 a good data annotator? In *Proceedings of the 61st annual meeting of the association for computational linguistics (acl)* (pp. 11173–11195). Association for Computational Linguistics.
- Ellingsen, J., Larsen, V. H., & Thorsrud, L. A. (2022). News media versus fred-md for macroeconomic forecasting. *Journal of Applied Econometrics*, 37(1), 63–81.
- Faust, J., & Wright, J. H. (2008). Efficient forecast tests for conditional policy forecasts. *Journal of Econometrics*, 146(2), 293–303.
- Garcia, D. (2013). Sentiment during recessions. *The Journal of Finance*, 68(3), 1267–1300.
- Genre, V., Kenny, G., Meyler, A., & Timmermann, A. (2013). Combining expert forecasts: Can anything beat the simple average? *International Journal of Forecasting*, 29(1), 108–121.
- Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3), 535–574.
- Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30).
- Gorodnichenko, Y., Pham, T., & Talavera, O. (2023). The voice of monetary policy. *American Economic Review*, 113(2), 548–584.
- Grootendorst, M. (2022). BERTopic: Neural Topic Modeling with a Class-based TF-IDF Procedure. *arXiv preprint arXiv:2203.05794*.
- Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don’t stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 8342–8360). Association for Computational Linguistics.
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Hansen, S., McMahon, M., & Prat, A. (2018). Transparency and deliberation within the FOMC: A computational linguistics approach. *The Quarterly Journal of Economics*, 133(2), 801–870.
- Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.
- Huang, A. H., Wang, H., & Yang, Y. (2023). FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2), 806–841.
- Johnson, S. G. B., & Tuckett, D. (2022). Narrative expectations in financial forecasting. *Journal of Behavioral Decision Making*, 35(1), 22–45.
- Ke, Z. T., Kelly, B., & Xiu, D. (2026). Enhancing investment decisions with sentiment analysis: A probabilistic ranking framework. *Journal of the American Statistical Association*, 1–15.
- Kim, H.-H., & Swanson, N. R. (2018). Methods for backcasting, nowcasting and forecasting using

- factor-MIDAS: With an application to Korean GDP. *Journal of Forecasting*, 37(3), 281–302.
- Lagerborg, A., Pappa, E., & Ravn, M. O. (2023). Sentimental Business Cycles. *The Review of Economic Studies*, 90(3), 1358–1393.
- Lawrence, M., Goodwin, P., O’Connor, M., & Onkal, D. (2006). Judgmental forecasting: A review of progress over the last 25 years. *International Journal of Forecasting*, 22(3), 493–518.
- Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., ... Zettlemoyer, L. (2020). BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 7871–7880). Association for Computational Linguistics.
- Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-ks. *The Journal of Finance*, 66(1), 35–65.
- Ludvigson, S. C. (2004). Consumer confidence and consumer spending. *Journal of Economic Perspectives*, 18(2), 29–50.
- Malo, P., Sinha, A., Korhonen, P., Wallenius, J., & Takala, P. (2014). Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- McCracken, M. W., & Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4), 574–589.
- Nordhaus, W. D. (1987). Forecasting efficiency: concepts and applications. *The Review of Economics and Statistics*, 667–674.
- Romer, C. D., & Romer, D. H. (2000). Federal Reserve information and the behavior of interest rates. *American Economic Review*, 90(3), 429–457.
- Rossi, B., & Sekhposyan, T. (2011, September). Understanding models’ forecasting performance. *Journal of Econometrics*, 164(1), 158–172.
- Röttger, P., & Pierrehumbert, J. (2021). Temporal adaptation of BERT and performance on downstream document classification: Insights from social media. In *Findings of the Association for Computational Linguistics: EMNLP 2021* (pp. 2400–2412). Association for Computational Linguistics.
- Rouzegar, H., & Makrehchi, M. (2024). Enhancing text classification through LLM-driven active learning and human annotation. In *Proceedings of the 18th Linguistic Annotation Workshop (LAW-XVIII)* (pp. 98–111). Association for Computational Linguistics.
- Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.
- Schorfheide, F., & Song, D. (2024). Real-time forecasting with a (standard) mixed-frequency VAR during a pandemic. *International Journal of Central Banking*, 20(4), 275–320.
- Schroeder, H., Roy, D., & Kabbara, J. (2025). Just put a human in the loop? Investigating LLM-assisted annotation for subjective tasks. In *Findings of the Association for Computational Linguistics: ACL 2025* (pp. 25771–25795). Association for Computational Linguistics.
- Shapiro, A. H., Sudhof, M., & Wilson, D. J. (2022). Measuring news sentiment. *Journal of*

- Econometrics*, 228(2), 221–243.
- Shapiro, L. (2011). *Embodied cognition*. Routledge/Taylor & Francis Group.
- Sharpe, S. A., Sinha, N. R., & Hollrah, C. A. (2023). The power of narrative sentiment in economic forecasts. *International Journal of Forecasting*, 39(3), 1097–1121.
- Slovic, P., Finucane, M., Peters, E., & MacGregor, D. G. (2002). Rational actors or rational fools: implications of the affect heuristic for behavioral economics. *The Journal of Socio-Economics*, 31(4), 329–342.
- Stark, T. (2013). SPF panelists’ forecasting methods: A note on the aggregate results of a November 2009 special survey. *Federal Reserve Bank of Philadelphia*.
- Stone, P. J., Dunphy, D. C., Smith, M. S., & Ogilvie, D. M. (1966). *The general inquirer: A computer approach to content analysis*. Cambridge, MA: MIT Press.
- Swanson, N. R. (1996). Forecasting using first-available versus fully revised economic time-series data. *Studies in Nonlinear Dynamics and Econometrics*, 1, 47–64.
- Swanson, N. R., Ghysels, E., & Callan, M. (1999). A multivariate time series analysis of the data revision process for industrial production and the composite leading indicator. In R. F. Engle & H. White (Eds.), *Cointegration, Causality, and Forecasting: Festschrift in honour of Clive W.J. Granger* (pp. 47–75). Oxford University Press.
- Swanson, N. R., & Van Dijk, D. (2006). Are statistical reporting agencies getting it right? Data rationality and business cycle asymmetry. *Journal of Business & Economic Statistics*, 24(1), 24–42.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3), 1139–1168.
- Tuckett, D., Smith, R., & Nyman, R. (2014). Tracking phantastic objects: A computer algorithmic investigation of narrative evolution in unstructured data sources. *Social Networks*, 38, 121–133.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems* (Vol. 30).
- Yin, W., Hay, J., & Roth, D. (2019). Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 3914–3923). Association for Computational Linguistics.

Table 1: Monthly Real-Time Variables Used in Econometric Model-Based Forecasting Experiments*

Forecast Target Variables	Abbrev.	Description	Data Transformation
Industrial Production	IP_t	Monthly real-time vintages of industrial production, measured in levels and seasonally adjusted. The index base year may vary across vintages.	$ip_t = \log(\frac{IP_t}{IP_{t-1}})$
CPI Inflation	CPI_t	Monthly real-time vintages of the consumer price index, measured in levels and seasonally adjusted. The index base year may vary across vintages.	$cpi_t = \log(\frac{CPI_t}{CPI_{t-1}})$
3-Month Treasury Bill Rate	$IR3M_t$	Monthly 3-month Treasury bill rate from FRED-MD. This series is treated as unrevised.	No transformation
10-Year Treasury Bond Rate	$IR10Y_t$	Monthly 10-year Treasury constant maturity yield from FRED-MD. This series is treated as unrevised.	No transformation
Unemployment Rate	UR_t	Monthly real-time vintages of the civilian unemployment rate, seasonally adjusted.	No transformation
Term Spread 1	$TS10Y_t$	Monthly term spread constructed from the 10-year Treasury yield and the 3-month Treasury bill rate.	$TS10Y_t = IR10Y_t - IR3M_t$
Term Spread 2	$TS1Y_t$	Monthly term spread constructed from the 1-year Treasury yield and the 3-month Treasury bill rate.	$TS1Y_t = IR1Y_t - IR3M_t$
S&P 500	$SP500_t$	Monthly return on the S&P 500 composite stock price index.	$sp500_t = \log(\frac{SP500_t}{SP500_{t-1}})$
VIX	VIX_t	Monthly CBOE Volatility Index.	$vix_t = \log(\frac{VIX_t}{VIX_{t-1}})$
Excess Bond Premium	EBP_t	Monthly excess bond premium of Gilchrist and Zakrajšek (2012).	No transformation
GZ Spread	GZ_t	Monthly corporate credit spread measure of Gilchrist and Zakrajšek (2012).	No transformation
Sentiment Variables	Abbrev.	Description	Data Transformation
FinBERT Sentiment	$S_{FB,t}$	Monthly narrative sentiment index constructed from macroeconomic and policy text using <i>FinBERT</i> , with document scores combined using source-balanced monthly aggregation.	No transformation
EconBERT Sentiment	$S_{EB,t}$	Monthly narrative sentiment index constructed from the same text corpus using a compact distilled transformer; monthly scores use source-balanced aggregation.	No transformation
Topic-Fused Sentiment	$S_{TF,t}$	Monthly topic-conditioned narrative sentiment index formed from topic membership weighted EconBERT tone within eight pre-specified macroeconomic topics and then equally weighted across the eight topic-specific tones.	No transformation
University of Michigan Consumer Sentiment Index	$S_{UM,t}$	Monthly survey-based consumer sentiment index from the University of Michigan.	No transformation
Conference Board Consumer Confidence	$S_{CB,t}$	Monthly survey-based consumer confidence index from the Conference Board.	No transformation

*Notes: This table lists the forecast target variables and sentiment variables used in the real-time forecasting experiments, along with the data transformation used for each variable. Macroeconomic variables are drawn from the monthly real-time FRED-MD database and related sources. The three narrative sentiment measures are constructed from macroeconomic and policy text and are designed to capture complementary dimensions of economic tone: a finance-domain measure (FinBERT), an economics-domain measure (EconBERT), and a topic-conditioned narrative measure (Topic-Fused). Sample period of all variables spans from 2003:1 - 2024:1. See Section 3 for complete details.

Table 2: Quarterly Real-Time Variables Used in Forecast Efficiency Experiments*

Variable	Abbrev.	Description	Transformation
Industrial Production	IP_t	Quarterly industrial production index, seasonally adjusted. The quarter t observation is constructed from the first calendar-month real-time vintage of quarter $t + 1$ and take the latest monthly industrial production observation available in that vintage.	$ip_t = 100[(\frac{IP_t}{IP_{t-1}})^4 - 1]$.
CPI	CPI_t	Quarterly consumer price index, seasonally adjusted. The quarter t observation is constructed from the first calendar-month real-time vintage of quarter $t + 1$ and take the latest monthly CPI observation available in that vintage.	$cpi_t = 100[(\frac{CPI_t}{CPI_{t-1}})^4 - 1]$.
3-Month Treasury Bill Rate	$IR3M_t$	Quarterly 3-month Treasury bill rate constructed from the monthly FRED-MD series. The quarter t observation is constructed from the first calendar-month real-time vintage of quarter $t + 1$ and take the latest monthly 3-month Treasury bill observation available in that vintage.	No transformation.
10-Year Treasury Bond Rate	$IR10Y_t$	Quarterly 10-year Treasury constant maturity yield constructed from the monthly FRED-MD series. The quarter t observation is constructed from the first calendar-month real-time vintage of quarter $t + 1$ and take the latest monthly 10-year Treasury yield observation available in that vintage.	No transformation.
Unemployment Rate	UR_t	Quarterly civilian unemployment rate, seasonally adjusted. The quarter t observation is constructed from the first calendar-month real-time vintage of quarter $t + 1$ and take the latest monthly unemployment rate observation available in that vintage.	No transformation.
FinBERT Sentiment	$S_{FB,t}$	Quarterly FinBERT narrative sentiment index, constructed from the monthly sentiment series defined in Table 1. The quarter t observation is the value for the first calendar month of quarter $t + 1$.	No transformation.
EconBERT Sentiment	$S_{EB,t}$	Quarterly EconBERT narrative sentiment index, constructed from the monthly sentiment series defined in Table 1. The quarter t observation is the value for the first calendar month of quarter $t + 1$.	No transformation.
Topic-Fused Sentiment	$S_{TF,t}$	Quarterly Topic-Fused narrative sentiment index, constructed from the monthly sentiment series defined in Table 1. The quarter t observation is the value for the first calendar month of quarter $t + 1$.	No transformation.
University of Michigan Consumer Sentiment Index	$S_{UM,t}$	Quarterly University of Michigan Consumer Sentiment Index, constructed from the monthly series defined in Table 1. The quarter t observation is the value for the first calendar month of quarter $t + 1$.	No transformation.
Conference Board Consumer Confidence	$S_{CB,t}$	Quarterly Conference Board Consumer Confidence Index, constructed from the monthly series defined in Table 1. The quarter t observation is the value for the first calendar month of quarter $t + 1$.	No transformation.

*Notes: This table lists the quarterly target variables and sentiment variables used in the SPF forecast efficiency experiments, along with the transformation applied to each variable. Here, t denotes the reference quarter immediately preceding the SPF survey, quarter $t + 1$ is therefore the survey quarter. Industrial production and CPI are constructed from monthly-vintaged real-time data containing monthly observations, while unemployment is the corresponding quarterly real-time vintage of THE survey quarter; these series are obtained from the Federal Reserve Bank of Philadelphia. The Treasury-rate variables are constructed from monthly FRED-MD series. For industrial production and CPI, we use the first calendar-month vintage of each SPF survey quarter and compare the latest available monthly observation with the observation three months earlier, taking both values from the same vintage. We then transformed to annualized quarter-over-quarter growth rates using the formula reported in the table. These transformations differ from the monthly log differences used in Table 1, and the resulting series match the frequency and units of the SPF forecasts. The five sentiment indices are defined in Table 1 and are converted to quarterly frequency using the first calendar-month observation of each SPF survey quarter. The sample period for all variables spans 2011:Q1–2024:Q1. See Section 3 for complete details.

Table 3: Forecasting Models Used in Econometric Model-Based Forecasting Experiments*

Model	Description
Benchmark: AR(SIC)	Autoregressive model with lags selected using the SIC.
Model 1: AR(SIC)+ S_1^{FB}	AR(SIC) model augmented with lagged FinBERT Sentiment.
Model 2: AR(SIC)+ S_1^{EB}	AR(SIC) model augmented with lagged EconBERT Sentiment.
Model 3: AR(SIC)+ S_1^{TF}	AR(SIC) model augmented with lagged Topic-Fused Sentiment.
Model 4: AR(SIC)+ S_1	AR(SIC) model augmented with all three lagged LLM based sentiment variables.
Model 5: AR(SIC)+ S_2^{CB}	AR(SIC) model augmented with lagged Conference Board Consumer Confidence Index.
Model 6: AR(SIC)+ S_2^{UM}	AR(SIC) model augmented with lagged University of Michigan Consumer Sentiment Index.
Model 7: AR(SIC)+ S_2	AR(SIC) model augmented with both consumer based sentiment variables.
Model 8: AR(SIC)+Macro	AR(SIC) model augmented with lagged macroeconomic variables.
Model 9: AR(SIC)+Macro+ S_1	AR(SIC) model augmented with lagged macroeconomic variables and three LLM based sentiment variables.
Model 10: AR(SIC)+Macro+ S_2	AR(SIC) model augmented with lagged macroeconomic variables, and two consumer based sentiment variables.
Model 11: AR(SIC)+Macro+ S_1+S_2	AR(SIC) model augmented with lagged macroeconomic variables, and all five sentiment variables.

*Notes: This table includes brief descriptions of the 11 different forecasting models used in monthly predictions of ip_t , cpi_t , $IR3M_t$, $IR10Y_t$, UR_t , $s\&p500_t$, vi_x_t , EBP_t , GZ_t , $TS1Y_t$, $TS10Y_t$. For complete details refer to Section 4.2.

Table 4: Summary of Econometric Model-Based Forecasting Experiment Results - Full Sample(2013:01–2024:01)*

h=1	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0150	1.0011	0.9981	1.0135	1.0364	0.9423***	0.9556**	0.9776**	0.9836	1.0033	1.0132
Model 2	YES	0.9790	1.0039	1.0184	0.9810	1.0578	1.0088	1.0020	1.0311	1.0408	1.0347	1.0322
Model 3	YES	0.9989	1.0282	1.0199	0.9849	1.0641	1.0076	0.9950	1.0402	1.0685	0.9994	1.0079
Model 4	YES	1.0103	1.0595	1.0555	0.9934	1.0285	0.9836	0.9889	1.0804	1.1256	1.0106	1.0526
Model 5	NO	1.0220	0.9954	1.0299	0.9834	1.0282	1.0000	1.0038	1.0030	1.0978	1.0196	1.0031
Model 6	NO	1.0319	0.9501	1.0480	0.9427	1.0206	1.0467	1.0109	1.0268	1.0619	1.1244	1.0229
Model 7	NO	1.0124	0.9480	1.1186	0.9195*	1.0600	1.0764	1.0245	1.0755	1.2489	1.1523	0.9920
Model 8	NO	1.2671	0.1383	0.6273	0.8861*	1.0622	1.0016	1.0549	1.1108	1.2533	1.0441	1.0035
Model 9	YES	1.2028	0.1401	0.6102	0.9233	1.0946	1.0472	1.0679	1.1920	1.3167	1.0345	1.0406
Model 10	NO	1.2758	0.1476	0.6655	0.9035*	1.1537	1.1353	1.1071	1.1872	1.4299	1.1083	1.2248
Model 11	YES	1.2941	0.1487	0.6393	0.9659	1.1244	1.1768	1.1189	1.2880	1.5521	1.0766	1.2008
h=3	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0121	0.9842	1.0126	1.0383	1.0536	0.9756	0.9983	0.9538**	1.0138	1.0098	1.0615
Model 2	YES	1.0225	1.0117	1.0349	0.9852	1.1416	1.0213	0.9995	1.0665	1.1054	1.0448	1.0415
Model 3	YES	1.0330	0.9774**	1.0191	0.9865	1.1317	1.0140	0.9983	1.0897	1.1783	1.0381	1.0693
Model 4	YES	0.9893	1.0241	1.0835	1.0106	1.1599	1.0462	1.0247	1.1879	1.3659	1.0999	1.0136
Model 5	NO	1.0364	1.0661	0.9642	0.9486	1.0409	1.0238	1.0098	0.9991	1.1061	1.0434	1.0015
Model 6	NO	1.0296	1.0406	0.9554	0.9271	0.9250	1.0698	1.0121	1.0440	1.2187	1.1059	1.1034
Model 7	NO	1.0089	1.0914	0.9808	0.8557*	0.9727	1.0973	1.0098	1.3059	1.7377	1.2751	1.0184
Model 8	NO	1.0419	1.2982	1.0269	1.1474	1.1402	1.0169	1.0060	1.6473	1.8324	1.0407	1.0212
Model 9	YES	1.0924	1.4648	1.1075	1.1845	1.2929	1.0667	1.0621	1.8708	2.0191	1.0801	1.0164
Model 10	NO	0.9685	1.8470	1.1015	1.0224	1.1734	1.0441	1.0420	1.6812	1.8174	1.2427	1.3717
Model 11	YES	1.0537	1.9924	1.1736	1.0668	1.2370	1.0402	1.0547	2.0658	2.2759	1.1830	1.5197
h=6	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0307	1.0297	1.0000	1.0953	1.1203	0.9896	1.0077	0.9780	1.0291	1.0134	0.9803
Model 2	YES	1.0314	1.0132	1.0224	0.9608	1.1685	1.0093	1.0059	1.0924	1.1055	1.0654	1.1168
Model 3	YES	1.0464	0.9938	1.0188	0.9664	1.2065	1.0088	1.0066	1.4163	1.4550	1.0344	1.1862
Model 4	YES	1.0446	1.0451	1.0152	0.9476**	1.2813	1.1211	1.0109	1.8385	2.0928	1.1172	1.0362
Model 5	NO	1.0104	1.0766	0.9570	1.0562	1.0642	1.0164	0.9973	1.1185	1.4084	1.0577	0.9713
Model 6	NO	1.0678	1.1111	0.9633	0.9426	0.8528	1.0260	0.9992	1.0523	1.2399	1.2802	1.2770
Model 7	NO	1.1647	1.0979	1.0380	0.8616**	1.0278	1.1739	1.0242	1.7579	2.4228	1.8804	1.0327
Model 8	NO	1.1682	1.2576	0.9746	1.1307	1.6155	1.0252	1.0463	1.9608	2.5297	1.0998	0.9350
Model 9	YES	1.1636	1.2425	0.9011	1.0696	2.0812	1.1025	1.0600	2.6401	3.0798	1.1902	0.9994
Model 10	NO	1.1650	1.4190	1.0621	1.0895	1.4386	1.1403	1.1790	1.9139	2.0527	1.5707	0.9001
Model 11	YES	1.1562	1.5110	0.9975	1.0241	1.7673	1.1620	1.1437	2.1193	2.1956	1.5942	0.9416
h=12	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0216	1.0700	1.0005	1.1264	1.1246	1.0075	1.0082	1.2434	1.0898	1.0124	0.9801
Model 2	YES	1.0058	0.9907	1.0076	1.0676	1.2377	1.0258	1.0028	1.1146	1.0782	1.0820	1.1180
Model 3	YES	1.0574	0.9112	1.0049	1.0655	1.3444	1.0127	1.0004	1.6187	1.3932	1.0081	1.2250
Model 4	YES	0.9285	0.9875	1.0314	1.0415	1.2705	1.1084	1.0316	1.3724	1.5144	1.1954	0.9998
Model 5	NO	1.0033	1.2573	0.9966	1.1315	1.0476	1.0632	1.0027	1.6553	2.4211	1.2003	0.7095**
Model 6	NO	1.0069	1.2206	0.9844	1.1544	0.8630	1.0447	0.9989	1.0129	1.4687	1.0891	1.2228
Model 7	NO	1.2482	1.4500	1.0333	0.9064	1.3734	1.1240	1.0164	2.0809	3.5353	1.7655	0.5966**
Model 8	NO	1.2708	1.0748	1.0787	0.9445	1.4469	1.2850	1.3696	2.2594	3.2672	1.0734	0.9568
Model 9	YES	1.1626	1.0323	1.0736	0.9310	1.5791	1.3981	1.4325	2.1358	3.1046	1.3227	0.9517
Model 10	NO	1.2061	1.2404	1.1612	1.0063	1.6134	1.4522	1.5162	1.7774	3.0778	1.0773	0.6399
Model 11	YES	1.1747	1.4003	1.2918	0.8692	1.6809	1.5473	1.6004	1.9268	3.2870	1.3732	0.5179**

*Notes: Numerical entries are relative MSFEs (ratio to the AR(SIC) benchmark). Minimum relative MSFEs for each variable and horizon are highlighted. Superscripts *, **, or *** denote rejection of the GW predictive accuracy null hypothesis, in favor of the alternative that the listed model has significantly lower MSFE than the AR(SIC) benchmark, at a 10 %, 5 %, or 1 % significance level, respectively, for MSFEs that are less than one. Starred entries associated with MSFEs greater than unity indicate that the AR(SIC) benchmark “wins”.

Table 5: Summary of Econometric Model-Based Forecasting Experiment Results - Pre-COVID Subsample (2013:01–2019:12)*

h=1	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0472	1.1200	1.0373	1.0348	1.0303	0.9838	0.9767*	1.0212	0.9670	0.9919	1.0090
Model 2	YES	1.0428	0.8840**	1.0499	1.0322	1.0412	0.9797	1.0015	1.1127	0.9769	1.1477	1.0319
Model 3	YES	1.0627	0.8590**	1.0278	1.0335	1.0172	0.9921	1.0026	1.2047	1.0802	1.1161	1.0341
Model 4	YES	1.0593	0.9760	1.0912	1.0548	0.9778	1.0419	1.0029	1.4103	1.2698	1.2005	1.0510
Model 5	NO	1.0531	0.9757	0.9963	0.8787**	1.0575	1.0273	1.0308	1.0346	1.2767	1.0841	0.9042*
Model 6	NO	1.0599	0.9205	1.0371	0.8984**	1.0755	0.9981	1.0177	1.0675	1.0441	1.1386	0.9215*
Model 7	NO	1.0566	0.9805	1.0829	0.8610**	1.1624	1.0846	1.0445	1.2828	1.5781	1.1599	0.9064
Model 8	NO	1.2723	0.9457	1.1027	0.9572	1.0883	1.0830	1.0703	1.6463	1.6059	1.2998	0.9531
Model 9	YES	1.3490	0.9850	1.0993	1.1661	1.0781	1.1583	1.0807	1.8497	1.6698	1.1650	1.0072
Model 10	NO	1.3270	0.8880	0.9476	0.9649	1.2227	1.2132	1.1326	1.6852	1.8406	1.4795	1.1348
Model 11	YES	1.4389	0.9457	0.9864	1.1162	1.1484	1.2119	1.1138	1.7986	2.0731	1.3631	1.2004
h=3	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0439	1.0590	1.0020	1.0080	1.0632	1.0177	1.0157	1.0029	1.0068	1.0476	1.1346
Model 2	YES	1.1396	0.5940***	1.0127	0.9492	1.0635	1.0410	1.0068	1.1940	1.0798	1.2257	1.1375
Model 3	YES	1.1388	0.5567***	1.0196	0.9673	1.0405	1.0300	1.0063	1.3308	1.3131	1.2383	1.2666
Model 4	YES	1.0284	0.6545***	1.0363	0.9538	1.1525	1.1699	1.0563	1.6257	1.6962	1.4326	1.1653
Model 5	NO	1.1019	0.9847	0.9964	0.7971*	1.1605	1.0780	1.0291	1.0214	1.2544	1.2835	0.7812*
Model 6	NO	1.1433	0.8270**	1.0614	0.8210*	1.1445	1.0712	1.0204	1.0487	1.2211	1.2695	0.8541
Model 7	NO	1.1342	1.0004	1.0900	0.7842*	1.2116	1.1235	1.0229	1.8682	2.4034	1.3588	0.7508**
Model 8	NO	1.0629	0.8145	1.0692	0.8842	1.1444	1.0635	1.0015	1.8726	2.6505	1.9868	0.7818**
Model 9	YES	1.0758	0.8217	1.0884	0.9185	1.3369	1.1883	1.0595	2.0701	2.6371	1.9082	0.8421**
Model 10	NO	1.1064	0.9285	0.8779**	0.8574	1.4359	1.0165	1.0364	1.6925	2.1184	1.8316	0.9918
Model 11	YES	1.1303	0.8346	0.8920*	0.8367	1.5690	1.1309	1.0979	1.8954	2.1379	1.8915	1.2277
h=6	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0362	1.0883	1.0063	1.0169	1.1299	0.9913	0.9891	0.9709	1.0239	1.0326	1.0254
Model 2	YES	1.0834	0.4362***	1.0134	0.9451	1.1605	1.0074	1.0047	1.2359	1.1117	1.1922	1.2251
Model 3	YES	1.1406	0.4336***	1.0537	0.9581	1.0931	1.0346	1.0071	1.8383	1.8209	1.2211	1.4314
Model 4	YES	1.0945	0.6555*	1.0683	0.9017	1.2747	1.2868	1.0101	2.6462	3.1179	1.2767	1.1603
Model 5	NO	1.0674	0.9796	1.0371	0.8037	1.2908	1.0699	1.0103	1.1426	1.8323	1.3285	0.5823**
Model 6	NO	1.1316	0.5383***	1.0232	0.7964	1.2383	1.0308	1.0071	0.9906	1.1811	1.2439	0.8137
Model 7	NO	1.1771	0.7244**	1.1809	0.7887	1.3814	1.3060	1.0267	2.2363	3.2847	1.4840	0.5155**
Model 8	NO	1.1387	1.1100	1.2030	0.9537	1.5863	1.1951	1.0242	2.3240	4.1373	2.2420	0.7968**
Model 9	YES	1.1269	1.0838	1.3059	0.9277	2.0132	1.2669	1.0114	3.6442	5.2192	2.3555	0.9556
Model 10	NO	1.1904	1.1390	1.1443	0.9262	1.9176	1.2505	1.0372	1.8532	2.6478	2.1827	0.6862*
Model 11	YES	1.1412	1.1551	1.1854	0.7766	2.0906	1.2683	1.0192	2.3466	2.8845	2.3607	0.7498
h=12	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0609	1.3663	0.9978	1.3092	1.0039	1.0145	0.9986	1.3131	1.0608	0.9974	1.0749
Model 2	YES	1.1521	0.4595***	1.0271	1.4130	1.0073	1.0095	0.9993	1.1829	1.0541	1.0269	1.1066
Model 3	YES	1.1657	0.4040***	1.0542	1.3333	1.0448	1.0233	1.0020	1.8803	1.5375	1.0510	1.4096
Model 4	YES	1.0213	0.5257***	1.1675	1.0604	1.1112	1.1961	1.0252	1.5126	1.6140	1.1510	0.9773
Model 5	NO	1.0498	0.8310	1.2034	1.3435	1.2893	1.1929	1.0121	1.8033	3.1285	1.5864	0.3827**
Model 6	NO	1.0908	0.4698**	1.0590	1.0809	1.2034	1.1838	1.0206	0.8593	1.5984	1.1864	0.5617**
Model 7	NO	1.1270	0.5691**	1.3719	1.2578	1.2852	1.2374	1.0123	2.4435	4.3665	1.9844	0.4089**
Model 8	NO	1.2762	0.7023	1.5140	1.3906	1.9410	1.3881	1.0282	2.2121	4.1857	1.5472	1.0047
Model 9	YES	1.3469	0.7476	1.5987	1.3656	1.7935	1.6839	1.0676	1.9482	3.7678	2.0326	1.0409
Model 10	NO	1.2241	0.6856*	1.5709	1.9534	2.0776	1.8371	1.0660	1.6318	3.4422	1.6022	0.8630
Model 11	YES	1.3071	0.5805**	1.5652	1.4021	1.9580	2.0090	1.1118	1.5407	3.3734	2.0968	0.5482*

*Notes: See details in Table 4

Table 6: Summary of Econometric Model-Based Forecasting Experiment Results - Post-2019 Sub-sample (2020:01–2024:01)*

h=1	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	0.9932	1.0009	0.9965	1.0100	1.0407	0.9214***	0.9315*	0.9671**	0.9923	1.0056	1.0158
Model 2	YES	0.9357	1.0041	1.0171	0.9724	1.0695	1.0234	1.0026	1.0113	1.0740	1.0120	1.0324
Model 3	YES	0.9557	1.0285	1.0196	0.9768	1.0970	1.0154	0.9863	1.0004	1.0625	0.9759	0.9922
Model 4	YES	0.9771	1.0596	1.0540	0.9832	1.0642	0.9544**	0.9728	1.0007	1.0508	0.9724	1.0536
Model 5	NO	1.0008	0.9954	1.0313	1.0008	1.0077	0.9863	0.9728*	0.9954	1.0049	1.0066	1.0624
Model 6	NO	1.0129	0.9501	1.0484	0.9501	0.9820	1.0711	1.0030	1.0169	1.0712	1.1216	1.0836
Model 7	NO	0.9824	0.9480	1.1200	0.9292	0.9880	1.0724	1.0015	1.0254	1.0780	1.1507	1.0433
Model 8	NO	1.2635	0.1370	0.6083	0.8743*	1.0438	0.9608	1.0373	0.9814	1.0702	0.9927	1.0336
Model 9	YES	1.1037	0.1387	0.5907	0.8829	1.1063	0.9916	1.0533	1.0330	1.1334	1.0083	1.0606
Model 10	NO	1.2410	0.1464	0.6542	0.8933*	1.1052	1.0963	1.0779	1.0667	1.2166	1.0337	1.2786
Model 11	YES	1.1959	0.1474	0.6254	0.9409	1.1075	1.1592	1.1247	1.1645	1.2815	1.0190	1.2010
h=3	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	0.9962	0.9829	1.0132	1.0497	1.0487	0.9539**	0.9784	0.9313***	1.0185	1.0039	1.0165
Model 2	YES	0.9641	1.0191	1.0362	0.9989	1.1809	1.0111	0.9912	1.0082	1.1224	1.0167	0.9825
Model 3	YES	0.9802	0.9848*	1.0191	0.9938	1.1775	1.0057	0.9891	0.9793	1.0885	1.0071	0.9478
Model 4	YES	0.9698	1.0306	1.0863	1.0322	1.1636	0.9826	0.9885	0.9875	1.1458	1.0483	0.9203
Model 5	NO	1.0038	1.0675	0.9623	1.0062	0.9808	0.9960	0.9877	0.9889	1.0073	1.0061	1.1372
Model 6	NO	0.9729	1.0444	0.9492	0.9674	0.8147	1.0690	1.0026	1.0418	1.2171	1.0805	1.2570
Model 7	NO	0.9464	1.0930	0.9744	0.8829	0.8526	1.0838	0.9948	1.0484	1.2941	1.2621	1.1832
Model 8	NO	1.0315	1.3067	1.0244	1.2474	1.1381	0.9930	1.0112	1.5441	1.2872	0.8939	1.1686
Model 9	YES	1.1006	1.4761	1.1086	1.2856	1.2708	1.0043	1.0651	1.7796	1.6073	0.9515	1.1238
Model 10	NO	0.8997	1.8632	1.1147	1.0850	1.0414	1.0582	1.0484	1.6761	1.6168	1.1512	1.6057
Model 11	YES	1.0155	2.0128	1.1902	1.1542	1.0701	0.9936	1.0050	2.1438	2.3678	1.0730	1.6996
h=6	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0281	1.0282	0.9996	1.1287	1.1162	0.9888	1.0294	0.9860	1.0344	1.0094	0.9505
Model 2	YES	1.0066	1.0286	1.0230	0.9674	1.1719	1.0103	1.0072	0.9306	1.0991	1.0386	1.0451
Model 3	YES	1.0014	1.0087	1.0166	0.9700*	1.2546	0.9962	1.0060	0.9404	1.0806	0.9950	1.0238
Model 4	YES	1.0209	1.0555	1.0119	0.9672	1.2841	1.0400	1.0117	0.9277	1.0441	1.0835	0.9540
Model 5	NO	0.9832**	1.0792	0.9519	1.1637	0.9681	0.9902	0.9822	1.0913	0.9747*	1.0005	1.2289
Model 6	NO	1.0374	1.1264	0.9595	1.0049	0.6893**	1.0237	0.9901	1.1218	1.3001	1.2879	1.5839
Model 7	NO	1.1587	1.1078	1.0289	0.8927*	0.8777	1.1092	1.0212	1.2183	1.5411	1.9640	1.3752
Model 8	NO	1.1823	1.2616	0.9600	1.2061	1.6279	0.9421	1.0720	1.5512	0.8851	0.8588	1.0265
Model 9	YES	1.1811	1.2467	0.8753	1.1300	2.1100	1.0221	1.1165	1.5077	0.8911	0.9444	1.0283
Model 10	NO	1.1529	1.4265	1.0568	1.1591	1.2354	1.0864	1.3442	1.9823	1.4438	1.4416	1.0417
Model 11	YES	1.1633	1.5205	0.9855	1.1294	1.6301	1.1100	1.2887	1.8630	1.4909	1.4326	1.0686
h=12	NAR	cpi_t	ur_t	ip_t	$IR3M_t$	$IR10Y_t$	$s\&p500_t$	vix_t	GZ_t	EBP_t	$TS1Y_t$	$TS10Y_t$
Model 1	YES	1.0027	1.0398	1.0008	1.0879	1.1661	1.0040	1.0194	1.0507	1.1451	1.0174	0.9075**
Model 2	YES	0.9352	1.0448	1.0059	0.9949	1.3169	1.0339	1.0069	0.9256	1.1241	1.1005	1.1268
Model 3	YES	1.0051	0.9628	1.0004	1.0091	1.4475	1.0075	0.9984	0.8951	1.1181	0.9936	1.0837
Model 4	YES	0.8837	1.0346	1.0191	1.0375	1.3252	1.0652	1.0390	0.9843	1.3245	1.2104	1.0170
Model 5	NO	0.9808**	1.3007	0.9778	1.0869	0.9646	0.9994	0.9917	1.2458	1.0721	1.0701	0.9595
Model 6	NO	0.9665	1.2971	0.9777	1.1698	0.7460*	0.9763*	0.9736**	1.4377	1.2213	1.0563	1.7287
Model 7	NO	1.3068	1.5397	1.0026	0.8324*	1.4037	1.0682	1.0211	1.0777	1.9502	1.6918	0.7402
Model 8	NO	1.2682	1.1127	1.0393	0.8506	1.2769	1.2343	1.7662	2.3902	1.5156	0.9136	0.9202
Model 9	YES	1.0736	1.0612	1.0260	0.8395	1.5054	1.2575	1.8563	2.6548	1.8399	1.0834	0.8835
Model 10	NO	1.1974	1.2968	1.1241	0.8069	1.4538	1.2628	2.0391	2.1802	2.3828	0.9003	0.4692**
Model 11	YES	1.1108	1.4838	1.2670	0.7570	1.5856	1.3202	2.1681	2.9949	3.1223	1.1292	0.4947*

*Notes: See details in Table 4.

Table 7: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.9295	0.6558	0.9341	0.8490
h=2	NO	NO	NO	NO	0.8916	0.6409	0.4171	0.8439
h=3	NO	NO	NO	NO	0.8626	0.5052	0.9576	0.8881
h=4	NO	NO	NO	NO	0.3340	0.8489	0.2473	0.9471
<i>CPI Inflation</i>								
h=1	YES	NO	NO	NO	0.0197	0.7511	0.4634	0.4162
h=2	YES	NO	NO	YES	0.0808	0.3653	0.6141	0.0350
h=3	YES	NO	NO	NO	0.0974	0.3030	0.5049	0.3549
h=4	YES	NO	NO	YES	0.0002	0.9515	0.3640	0.0888
<i>3-Month Treasury Bill Rate</i>								
h=1	NO	NO	NO	NO	0.3380	0.8321	0.3287	0.1227
h=2	NO	NO	YES	YES	0.2165	0.7075	0.0856	0.0812
h=3	YES	NO	YES	NO	0.0842	0.8769	0.0415	0.1051
h=4	YES	NO	NO	YES	0.0175	0.8139	0.1107	0.0320
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0238	0.8353	0.6297	0.0747
h=2	YES	NO	NO	YES	0.0001	0.6502	0.1677	0.0005
h=3	YES	NO	NO	YES	0.0003	0.9589	0.7825	0.0079
h=4	YES	NO	NO	YES	0.0004	0.8336	0.6598	0.0106
<i>Unemployment</i>								
h=1	YES	YES	YES	YES	0.0000	0.0442	0.0977	0.0000
h=2	YES	NO	NO	YES	0.0028	0.8543	0.1266	0.0012
h=3	YES	NO	NO	YES	0.0005	0.5490	0.1090	0.0155
h=4	YES	NO	YES	NO	0.0135	0.3301	0.0480	0.1072

*Notes: The first four test-result columns report whether the corresponding null hypothesis is rejected, and the final four columns report the associated p-values. The hypotheses are (1) $H_0 : \alpha = 0, \beta = 0, \gamma = 0$; (2) $H_0 : \alpha = 0$; (3) $H_0 : \beta = 0$; and (4) $H_0 : \gamma = 0$. The p-value for (2) is the HAC-robust coefficient t-test p-value for the intercept. The p-values for (1), (3), and (4) are from robust Wald F-tests using the HAC covariance matrix. Here, β denotes the vector of coefficients on the three LLM-based sentiment indices (FinBERT, EconBERT, and Topic-Fused), and γ denotes the vector of coefficients on the five macroeconomic variables. A bold YES denotes rejection at the 10% level; the associated p-values (below 0.10) are likewise shown in bold. Standard errors are HAC (Newey–West) with lag truncation equal to manuscript horizon h . Sample: 2011Q1–2024Q1.

Table 8: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.7371	0.2796	0.3630	0.8097
h=2	NO	NO	NO	NO	0.6528	0.6411	0.2513	0.5994
h=3	NO	NO	NO	NO	0.9010	0.9335	0.6907	0.8922
h=4	NO	NO	NO	NO	0.9126	0.7661	0.6887	0.8325
<i>CPI Inflation</i>								
h=1	YES	NO	YES	YES	0.0091	0.3606	0.0826	0.0016
h=2	NO	NO	NO	YES	0.1030	0.5758	0.3797	0.0779
h=3	NO	NO	YES	YES	0.1006	0.2405	0.0498	0.0825
h=4	YES	NO	YES	YES	0.0000	0.3940	0.0992	0.0000
<i>3-Month Treasury Bill Rate</i>								
h=1	NO	YES	NO	YES	0.1069	0.0431	0.1297	0.0398
h=2	YES	YES	YES	YES	0.0000	0.0185	0.0064	0.0004
h=3	YES	YES	YES	YES	0.0000	0.0040	0.0002	0.0007
h=4	YES	YES	YES	YES	0.0000	0.0021	0.0004	0.0000
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	YES	YES	YES	0.0021	0.0049	0.0630	0.0091
h=2	YES	YES	YES	YES	0.0003	0.0142	0.0416	0.0009
h=3	YES	YES	YES	YES	0.0001	0.0135	0.0199	0.0082
h=4	YES	YES	YES	YES	0.0000	0.0014	0.0142	0.0315
<i>Unemployment</i>								
h=1	YES	YES	YES	YES	0.0000	0.0985	0.0262	0.0000
h=2	YES	NO	NO	YES	0.0000	0.2790	0.1323	0.0000
h=3	YES	NO	YES	YES	0.0000	0.4685	0.0900	0.0001
h=4	NO	NO	NO	NO	0.2810	0.2624	0.1687	0.1714

*Notes: The first four test-result columns report whether the corresponding null hypothesis is rejected, and the final four columns report the associated p-values. The hypotheses are (1) $H_0 : \alpha = 0, \beta = 0, \gamma = 0$; (2) $H_0 : \alpha = 0$; (3) $H_0 : \beta = 0$; and (4) $H_0 : \gamma = 0$. The p-value for (2) is the HAC-robust coefficient t-test p-value for the intercept. The p-values for (1), (3), and (4) are from robust Wald F-tests using the HAC covariance matrix. Here, β denotes the vector of coefficients on the University of Michigan Consumer Sentiment Index and the Conference Board Consumer Confidence Index, and γ denotes the vector of coefficients on the five macroeconomic variables. A YES denotes rejection at the 10% level; p-values below 0.10 are shown in bold. Standard errors are HAC (Newey–West) with lag truncation equal to manuscript horizon h . Sample: 2011Q1–2024Q1.

Figure 1: LLM-based narrative sentiment and the eleven forecasting targets, 2013:1 - 2024:1. FinBERT, EconBERT, Topic-Fused, and the target series are scaled separately within each panel.

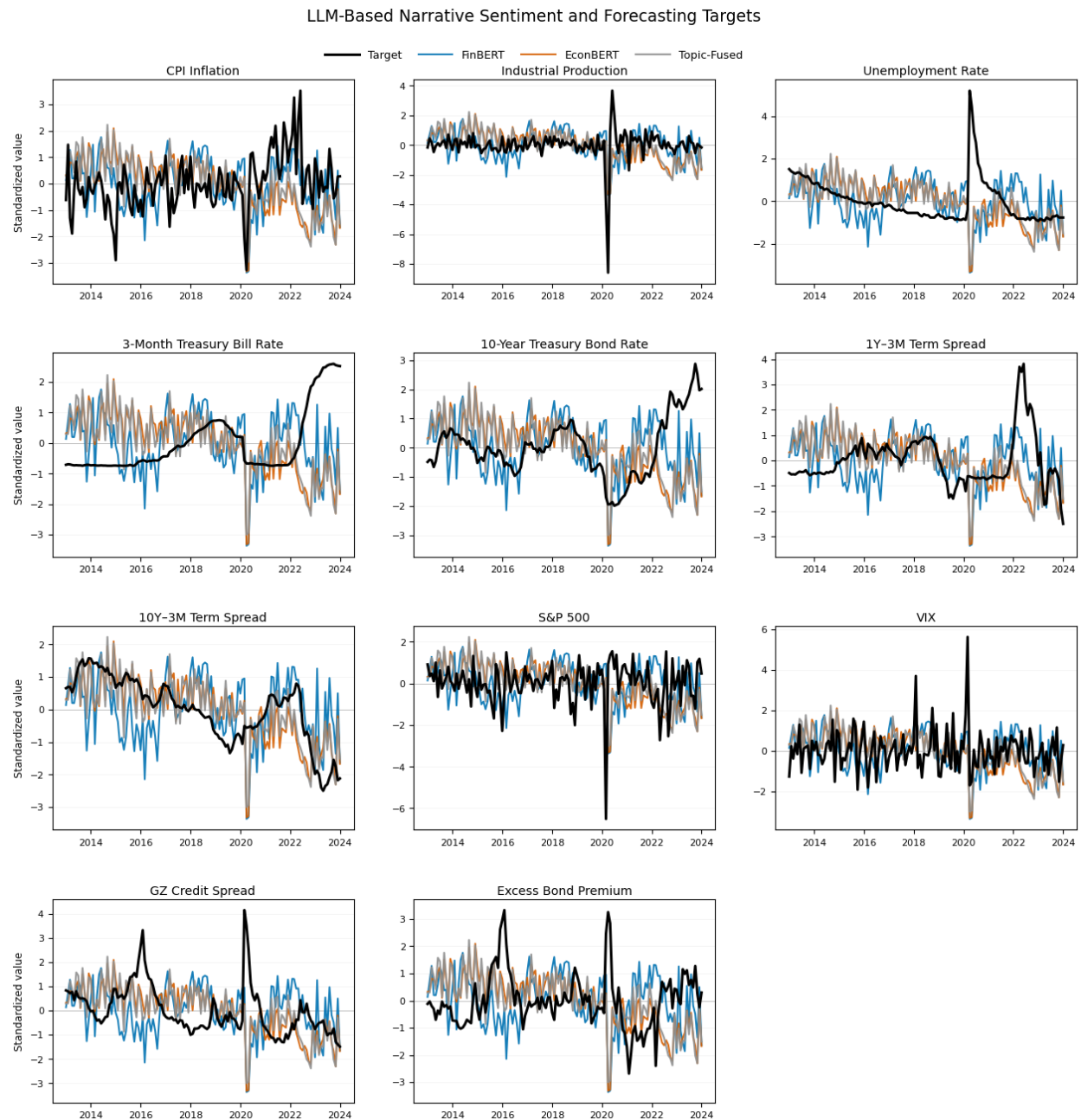


Figure 2: Survey-based sentiment and the eleven forecasting targets, 2013:1 - 2024:1. The University of Michigan index, Conference Board index, and the target series are scaled separately within each panel.

